

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION  
CSE 4316: SENIOR DESIGN I  
FALL 2024**



**EXCALIBUR  
LEARNLEAF ORGANIZER**

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## CONTENTS

|          |                                              |           |
|----------|----------------------------------------------|-----------|
| <b>1</b> | <b>Product Concept</b>                       | <b>25</b> |
| 1.1      | Purpose and Use . . . . .                    | 25        |
| 1.2      | Intended Audience . . . . .                  | 25        |
| <b>2</b> | <b>Product Description</b>                   | <b>26</b> |
| 2.1      | Features & Functions . . . . .               | 26        |
| 2.2      | External Inputs & Outputs . . . . .          | 27        |
| 2.3      | Product Interfaces . . . . .                 | 27        |
| <b>3</b> | <b>Customer Requirements</b>                 | <b>29</b> |
| 3.1      | Service Worker for Offline Support . . . . . | 29        |
| 3.1.1    | Description . . . . .                        | 29        |
| 3.1.2    | Source . . . . .                             | 29        |
| 3.1.3    | Constraints . . . . .                        | 29        |
| 3.1.4    | Standards . . . . .                          | 29        |
| 3.1.5    | Priority . . . . .                           | 29        |
| 3.2      | Add to Home Screen Feature . . . . .         | 29        |
| 3.2.1    | Description . . . . .                        | 29        |
| 3.2.2    | Source . . . . .                             | 29        |
| 3.2.3    | Constraints . . . . .                        | 29        |
| 3.2.4    | Standards . . . . .                          | 29        |
| 3.2.5    | Priority . . . . .                           | 29        |
| 3.3      | Push Notifications for Reminders . . . . .   | 29        |
| 3.3.1    | Description . . . . .                        | 29        |
| 3.3.2    | Source . . . . .                             | 30        |
| 3.3.3    | Constraints . . . . .                        | 30        |
| 3.3.4    | Standards . . . . .                          | 30        |
| 3.3.5    | Priority . . . . .                           | 30        |
| 3.4      | Mobile-First Responsive Design . . . . .     | 30        |
| 3.4.1    | Description . . . . .                        | 30        |
| 3.4.2    | Source . . . . .                             | 30        |
| 3.4.3    | Constraints . . . . .                        | 30        |
| 3.4.4    | Standards . . . . .                          | 30        |
| 3.4.5    | Priority . . . . .                           | 30        |
| 3.5      | Data Sync Across Devices . . . . .           | 30        |
| 3.5.1    | Description . . . . .                        | 30        |
| 3.5.2    | Source . . . . .                             | 30        |
| 3.5.3    | Constraints . . . . .                        | 30        |
| 3.5.4    | Standards . . . . .                          | 30        |
| 3.5.5    | Priority . . . . .                           | 30        |
| 3.6      | Manual Refreshing . . . . .                  | 31        |
| 3.6.1    | Description . . . . .                        | 31        |
| 3.6.2    | Source . . . . .                             | 31        |
| 3.6.3    | Constraints . . . . .                        | 31        |
| 3.6.4    | Standards . . . . .                          | 31        |
| 3.6.5    | Priority . . . . .                           | 31        |

|        |                                        |    |
|--------|----------------------------------------|----|
| 3.7    | Set Notification Preferences . . . . . | 31 |
| 3.7.1  | Description . . . . .                  | 31 |
| 3.7.2  | Source . . . . .                       | 31 |
| 3.7.3  | Constraints . . . . .                  | 31 |
| 3.7.4  | Standards . . . . .                    | 31 |
| 3.7.5  | Priority . . . . .                     | 31 |
| 3.8    | Updated Preference Message . . . . .   | 31 |
| 3.8.1  | Description . . . . .                  | 31 |
| 3.8.2  | Source . . . . .                       | 31 |
| 3.8.3  | Constraints . . . . .                  | 31 |
| 3.8.4  | Standards . . . . .                    | 32 |
| 3.8.5  | Priority . . . . .                     | 32 |
| 3.9    | Weekly Notifications . . . . .         | 32 |
| 3.9.1  | Description . . . . .                  | 32 |
| 3.9.2  | Source . . . . .                       | 32 |
| 3.9.3  | Constraints . . . . .                  | 32 |
| 3.9.4  | Standards . . . . .                    | 32 |
| 3.9.5  | Priority . . . . .                     | 32 |
| 3.10   | Daily Notifications . . . . .          | 32 |
| 3.10.1 | Description . . . . .                  | 32 |
| 3.10.2 | Source . . . . .                       | 32 |
| 3.10.3 | Constraints . . . . .                  | 32 |
| 3.10.4 | Standards . . . . .                    | 32 |
| 3.10.5 | Priority . . . . .                     | 32 |
| 3.11   | Urgent Notifications . . . . .         | 32 |
| 3.11.1 | Description . . . . .                  | 32 |
| 3.11.2 | Source . . . . .                       | 32 |
| 3.11.3 | Constraints . . . . .                  | 33 |
| 3.11.4 | Standards . . . . .                    | 33 |
| 3.11.5 | Priority . . . . .                     | 33 |
| 3.12   | Notification Content . . . . .         | 33 |
| 3.12.1 | Description . . . . .                  | 33 |
| 3.12.2 | Source . . . . .                       | 33 |
| 3.12.3 | Constraints . . . . .                  | 33 |
| 3.12.4 | Standards . . . . .                    | 33 |
| 3.12.5 | Priority . . . . .                     | 33 |
| 3.13   | Notification Formatting . . . . .      | 33 |
| 3.13.1 | Description . . . . .                  | 33 |
| 3.13.2 | Source . . . . .                       | 33 |
| 3.13.3 | Constraints . . . . .                  | 33 |
| 3.13.4 | Standards . . . . .                    | 33 |
| 3.13.5 | Priority . . . . .                     | 33 |
| 3.14   | Project Page Theme . . . . .           | 34 |
| 3.14.1 | Description . . . . .                  | 34 |
| 3.14.2 | Source . . . . .                       | 34 |
| 3.14.3 | Constraints . . . . .                  | 34 |
| 3.14.4 | Standards . . . . .                    | 34 |
| 3.14.5 | Priority . . . . .                     | 34 |

|        |                               |    |
|--------|-------------------------------|----|
| 3.15   | Project Widget Theme          | 34 |
| 3.15.1 | Description                   | 34 |
| 3.15.2 | Source                        | 34 |
| 3.15.3 | Constraints                   | 34 |
| 3.15.4 | Standards                     | 34 |
| 3.15.5 | Priority                      | 34 |
| 3.16   | Project Widget Doughnut Chart | 34 |
| 3.16.1 | Description                   | 34 |
| 3.16.2 | Source                        | 34 |
| 3.16.3 | Constraints                   | 35 |
| 3.16.4 | Standards                     | 35 |
| 3.16.5 | Priority                      | 35 |
| 3.17   | Project Tasks                 | 35 |
| 3.17.1 | Description                   | 35 |
| 3.17.2 | Source                        | 35 |
| 3.17.3 | Constraints                   | 35 |
| 3.17.4 | Standards                     | 35 |
| 3.17.5 | Priority                      | 35 |
| 3.18   | Project Deletion              | 35 |
| 3.18.1 | Description                   | 35 |
| 3.18.2 | Source                        | 35 |
| 3.18.3 | Constraints                   | 35 |
| 3.18.4 | Standards                     | 35 |
| 3.18.5 | Priority                      | 35 |
| 3.19   | Project Delete Button Theme   | 36 |
| 3.19.1 | Description                   | 36 |
| 3.19.2 | Source                        | 36 |
| 3.19.3 | Constraints                   | 36 |
| 3.19.4 | Standards                     | 36 |
| 3.19.5 | Priority                      | 36 |
| 3.20   | Project Widget Tooltip        | 36 |
| 3.20.1 | Description                   | 36 |
| 3.20.2 | Source                        | 36 |
| 3.20.3 | Constraints                   | 36 |
| 3.20.4 | Standards                     | 36 |
| 3.20.5 | Priority                      | 36 |
| 3.21   | Project Widget Buttons        | 36 |
| 3.21.1 | Description                   | 36 |
| 3.21.2 | Source                        | 36 |
| 3.21.3 | Constraints                   | 36 |
| 3.21.4 | Standards                     | 36 |
| 3.21.5 | Priority                      | 36 |
| 3.22   | Filter Projects by Due Date   | 37 |
| 3.22.1 | Description                   | 37 |
| 3.22.2 | Source                        | 37 |
| 3.22.3 | Constraints                   | 37 |
| 3.22.4 | Standards                     | 37 |
| 3.22.5 | Priority                      | 37 |

|        |                                        |    |
|--------|----------------------------------------|----|
| 3.23   | Filter Projects by Subject . . . . .   | 37 |
| 3.23.1 | Description . . . . .                  | 37 |
| 3.23.2 | Source . . . . .                       | 37 |
| 3.23.3 | Constraints . . . . .                  | 37 |
| 3.23.4 | Standards . . . . .                    | 37 |
| 3.23.5 | Priority . . . . .                     | 37 |
| 3.24   | Search for Projects by Name . . . . .  | 37 |
| 3.24.1 | Description . . . . .                  | 37 |
| 3.24.2 | Source . . . . .                       | 37 |
| 3.24.3 | Constraints . . . . .                  | 37 |
| 3.24.4 | Standards . . . . .                    | 37 |
| 3.24.5 | Priority . . . . .                     | 37 |
| 3.25   | FAB Theme . . . . .                    | 38 |
| 3.25.1 | Description . . . . .                  | 38 |
| 3.25.2 | Source . . . . .                       | 38 |
| 3.25.3 | Constraints . . . . .                  | 38 |
| 3.25.4 | Standards . . . . .                    | 38 |
| 3.25.5 | Priority . . . . .                     | 38 |
| 3.26   | Adding Tasks to Project . . . . .      | 38 |
| 3.26.1 | Description . . . . .                  | 38 |
| 3.26.2 | Source . . . . .                       | 38 |
| 3.26.3 | Constraints . . . . .                  | 38 |
| 3.26.4 | Standards . . . . .                    | 38 |
| 3.26.5 | Priority . . . . .                     | 38 |
| 3.27   | Project Archive Button Theme . . . . . | 38 |
| 3.27.1 | Description . . . . .                  | 38 |
| 3.27.2 | Source . . . . .                       | 38 |
| 3.27.3 | Constraints . . . . .                  | 38 |
| 3.27.4 | Standards . . . . .                    | 38 |
| 3.27.5 | Priority . . . . .                     | 39 |
| 3.28   | Project Widget Content . . . . .       | 39 |
| 3.28.1 | Description . . . . .                  | 39 |
| 3.28.2 | Source . . . . .                       | 39 |
| 3.28.3 | Constraints . . . . .                  | 39 |
| 3.28.4 | Standards . . . . .                    | 39 |
| 3.28.5 | Priority . . . . .                     | 39 |
| 3.29   | Project Deletion . . . . .             | 39 |
| 3.29.1 | Description . . . . .                  | 39 |
| 3.29.2 | Source . . . . .                       | 39 |
| 3.29.3 | Constraints . . . . .                  | 39 |
| 3.29.4 | Standards . . . . .                    | 39 |
| 3.29.5 | Priority . . . . .                     | 39 |
| 3.30   | Project Task Management . . . . .      | 39 |
| 3.30.1 | Description . . . . .                  | 39 |
| 3.30.2 | Source . . . . .                       | 39 |
| 3.30.3 | Constraints . . . . .                  | 39 |
| 3.30.4 | Standards . . . . .                    | 39 |
| 3.30.5 | Priority . . . . .                     | 40 |

|        |                                   |    |
|--------|-----------------------------------|----|
| 3.31   | Project Edit Button Theme         | 40 |
| 3.31.1 | Description                       | 40 |
| 3.31.2 | Source                            | 40 |
| 3.31.3 | Constraints                       | 40 |
| 3.31.4 | Standards                         | 40 |
| 3.31.5 | Priority                          | 40 |
| 3.32   | Project Manual Reordering         | 40 |
| 3.32.1 | Description                       | 40 |
| 3.32.2 | Source                            | 40 |
| 3.32.3 | Constraints                       | 40 |
| 3.32.4 | Standards                         | 40 |
| 3.32.5 | Priority                          | 40 |
| 3.33   | Project Dashboard Dynamic Display | 40 |
| 3.33.1 | Description                       | 40 |
| 3.33.2 | Source                            | 40 |
| 3.33.3 | Constraints                       | 41 |
| 3.33.4 | Standards                         | 41 |
| 3.33.5 | Priority                          | 41 |
| 3.34   | Remember-Me Functionality         | 41 |
| 3.34.1 | Source                            | 41 |
| 3.34.2 | Description                       | 41 |
| 3.34.3 | Requirements Area                 | 41 |
| 3.34.4 | Importance                        | 41 |
| 3.34.5 | Applicable Standard               | 41 |
| 3.34.6 | Constraints                       | 41 |
| 3.35   | Error Handling and Feedback       | 41 |
| 3.35.1 | Source                            | 41 |
| 3.35.2 | Description                       | 41 |
| 3.35.3 | Requirements Area                 | 41 |
| 3.35.4 | Importance                        | 41 |
| 3.35.5 | Applicable Standard               | 41 |
| 3.35.6 | Constraints                       | 41 |
| 3.36   | Responsive Design for Login Page  | 42 |
| 3.36.1 | Source                            | 42 |
| 3.36.2 | Description                       | 42 |
| 3.36.3 | Requirements Area                 | 42 |
| 3.36.4 | Importance                        | 42 |
| 3.36.5 | Applicable Standard               | 42 |
| 3.36.6 | Constraints                       | 42 |
| 3.37   | User-Friendly Registration Form   | 42 |
| 3.37.1 | Source                            | 42 |
| 3.37.2 | Description                       | 42 |
| 3.37.3 | Requirements Area                 | 42 |
| 3.37.4 | Importance                        | 42 |
| 3.37.5 | Applicable Standard               | 42 |
| 3.37.6 | Constraints                       | 42 |
| 3.38   | Terms of Service Agreement        | 42 |
| 3.38.1 | Source                            | 42 |

|        |                                       |    |
|--------|---------------------------------------|----|
| 3.38.2 | Description . . . . .                 | 42 |
| 3.38.3 | Requirements Area . . . . .           | 42 |
| 3.38.4 | Importance . . . . .                  | 42 |
| 3.38.5 | Applicable Standard . . . . .         | 43 |
| 3.38.6 | Constraints . . . . .                 | 43 |
| 3.39   | Data Privacy Notice . . . . .         | 43 |
| 3.39.1 | Source . . . . .                      | 43 |
| 3.39.2 | Description . . . . .                 | 43 |
| 3.39.3 | Requirements Area . . . . .           | 43 |
| 3.39.4 | Importance . . . . .                  | 43 |
| 3.39.5 | Applicable Standard . . . . .         | 43 |
| 3.39.6 | Constraints . . . . .                 | 43 |
| 3.40   | Social Media Sign-up Option . . . . . | 43 |
| 3.40.1 | Source . . . . .                      | 43 |
| 3.40.2 | Description . . . . .                 | 43 |
| 3.40.3 | Requirements Area . . . . .           | 43 |
| 3.40.4 | Importance . . . . .                  | 43 |
| 3.40.5 | Applicable Standard . . . . .         | 43 |
| 3.40.6 | Constraints . . . . .                 | 43 |
| 3.41   | Progressive Form Filling . . . . .    | 43 |
| 3.41.1 | Source . . . . .                      | 43 |
| 3.41.2 | Description . . . . .                 | 43 |
| 3.41.3 | Requirements Area . . . . .           | 44 |
| 3.41.4 | Importance . . . . .                  | 44 |
| 3.41.5 | Applicable Standard . . . . .         | 44 |
| 3.41.6 | Constraints . . . . .                 | 44 |
| 3.42   | Push Notification System . . . . .    | 44 |
| 3.42.1 | Source . . . . .                      | 44 |
| 3.42.2 | Description . . . . .                 | 44 |
| 3.42.3 | Requirements Area . . . . .           | 44 |
| 3.42.4 | Importance . . . . .                  | 44 |
| 3.42.5 | Applicable Standard . . . . .         | 44 |
| 3.42.6 | Constraints . . . . .                 | 44 |
| 3.43   | Biometric Authentication . . . . .    | 44 |
| 3.43.1 | Source . . . . .                      | 44 |
| 3.43.2 | Description . . . . .                 | 44 |
| 3.43.3 | Requirements Area . . . . .           | 44 |
| 3.43.4 | Importance . . . . .                  | 44 |
| 3.43.5 | Applicable Standard . . . . .         | 44 |
| 3.43.6 | Constraints . . . . .                 | 45 |
| 3.44   | Camera Integration . . . . .          | 45 |
| 3.44.1 | Source . . . . .                      | 45 |
| 3.44.2 | Description . . . . .                 | 45 |
| 3.44.3 | Requirements Area . . . . .           | 45 |
| 3.44.4 | Importance . . . . .                  | 45 |
| 3.44.5 | Applicable Standard . . . . .         | 45 |
| 3.44.6 | Constraints . . . . .                 | 45 |
| 3.45   | Deep Linking . . . . .                | 45 |

|        |                                      |    |
|--------|--------------------------------------|----|
| 3.45.1 | Source . . . . .                     | 45 |
| 3.45.2 | Description . . . . .                | 45 |
| 3.45.3 | Requirements Area . . . . .          | 45 |
| 3.45.4 | Importance . . . . .                 | 45 |
| 3.45.5 | Applicable Standard . . . . .        | 45 |
| 3.45.6 | Constraints . . . . .                | 45 |
| 3.46   | Haptic Feedback . . . . .            | 45 |
| 3.46.1 | Source . . . . .                     | 45 |
| 3.46.2 | Description . . . . .                | 45 |
| 3.46.3 | Requirements Area . . . . .          | 46 |
| 3.46.4 | Importance . . . . .                 | 46 |
| 3.46.5 | Applicable Standard . . . . .        | 46 |
| 3.46.6 | Constraints . . . . .                | 46 |
| 3.47   | Push Notification Channels . . . . . | 46 |
| 3.47.1 | Source . . . . .                     | 46 |
| 3.47.2 | Description . . . . .                | 46 |
| 3.47.3 | Requirements Area . . . . .          | 46 |
| 3.47.4 | Importance . . . . .                 | 46 |
| 3.47.5 | Applicable Standard . . . . .        | 46 |
| 3.47.6 | Constraints . . . . .                | 46 |
| 3.48   | Mobile-specific UI/UX . . . . .      | 46 |
| 3.48.1 | Source . . . . .                     | 46 |
| 3.48.2 | Description . . . . .                | 46 |
| 3.48.3 | Requirements Area . . . . .          | 46 |
| 3.48.4 | Importance . . . . .                 | 46 |
| 3.48.5 | Applicable Standard . . . . .        | 46 |
| 3.48.6 | Constraints . . . . .                | 46 |
| 3.49   | Reactivate Archived Items . . . . .  | 47 |
| 3.49.1 | Source . . . . .                     | 47 |
| 3.49.2 | Description . . . . .                | 47 |
| 3.49.3 | Requirements Area . . . . .          | 47 |
| 3.49.4 | Importance . . . . .                 | 47 |
| 3.49.5 | Applicable Standard . . . . .        | 47 |
| 3.49.6 | Constraints . . . . .                | 47 |
| 3.50   | Edit Archived Items . . . . .        | 47 |
| 3.50.1 | Source . . . . .                     | 47 |
| 3.50.2 | Description . . . . .                | 47 |
| 3.50.3 | Requirements Area . . . . .          | 47 |
| 3.50.4 | Importance . . . . .                 | 47 |
| 3.50.5 | Applicable Standard . . . . .        | 47 |
| 3.50.6 | Constraints . . . . .                | 47 |
| 3.51   | Delete Archived Items . . . . .      | 47 |
| 3.51.1 | Source . . . . .                     | 47 |
| 3.51.2 | Description . . . . .                | 47 |
| 3.51.3 | Requirements Area . . . . .          | 47 |
| 3.51.4 | Importance . . . . .                 | 47 |
| 3.51.5 | Applicable Standard . . . . .        | 48 |
| 3.51.6 | Constraints . . . . .                | 48 |

|        |                                     |    |
|--------|-------------------------------------|----|
| 3.52   | Toggle Archive Visibility . . . . . | 48 |
| 3.52.1 | Source . . . . .                    | 48 |
| 3.52.2 | Description . . . . .               | 48 |
| 3.52.3 | Requirements Area . . . . .         | 48 |
| 3.52.4 | Importance . . . . .                | 48 |
| 3.52.5 | Applicable Standard . . . . .       | 48 |
| 3.52.6 | Constraints . . . . .               | 48 |
| 3.53   | Filter Tasks . . . . .              | 48 |
| 3.53.1 | Source . . . . .                    | 48 |
| 3.53.2 | Description . . . . .               | 48 |
| 3.53.3 | Requirements Area . . . . .         | 48 |
| 3.53.4 | Importance . . . . .                | 48 |
| 3.53.5 | Applicable Standard . . . . .       | 48 |
| 3.53.6 | Constraints . . . . .               | 48 |
| 3.54   | Task Page Title . . . . .           | 48 |
| 3.54.1 | Source . . . . .                    | 48 |
| 3.54.2 | Description . . . . .               | 49 |
| 3.54.3 | Requirements Area . . . . .         | 49 |
| 3.54.4 | Importance . . . . .                | 49 |
| 3.54.5 | Applicable Standard . . . . .       | 49 |
| 3.54.6 | Constraints . . . . .               | 49 |
| 3.55   | New Task Menu Button . . . . .      | 49 |
| 3.55.1 | Source . . . . .                    | 49 |
| 3.55.2 | Description . . . . .               | 49 |
| 3.55.3 | Requirements Area . . . . .         | 49 |
| 3.55.4 | Importance . . . . .                | 49 |
| 3.55.5 | Applicable Standard . . . . .       | 49 |
| 3.55.6 | Constraints . . . . .               | 49 |
| 3.56   | Task Title Input . . . . .          | 49 |
| 3.56.1 | Source . . . . .                    | 49 |
| 3.56.2 | Description . . . . .               | 49 |
| 3.56.3 | Requirements Area . . . . .         | 49 |
| 3.56.4 | Importance . . . . .                | 49 |
| 3.56.5 | Applicable Standard . . . . .       | 50 |
| 3.56.6 | Constraints . . . . .               | 50 |
| 3.57   | Task Description Input . . . . .    | 50 |
| 3.57.1 | Source . . . . .                    | 50 |
| 3.57.2 | Description . . . . .               | 50 |
| 3.57.3 | Requirements Area . . . . .         | 50 |
| 3.57.4 | Importance . . . . .                | 50 |
| 3.57.5 | Applicable Standard . . . . .       | 50 |
| 3.57.6 | Constraints . . . . .               | 50 |
| 3.58   | Task Subject Selection . . . . .    | 50 |
| 3.58.1 | Source . . . . .                    | 50 |
| 3.58.2 | Description . . . . .               | 50 |
| 3.58.3 | Requirements Area . . . . .         | 50 |
| 3.58.4 | Importance . . . . .                | 50 |
| 3.58.5 | Applicable Standard . . . . .       | 50 |

|                                          |    |
|------------------------------------------|----|
| 3.58.6 Constraints . . . . .             | 50 |
| 3.59 Task Project Selection . . . . .    | 51 |
| 3.59.1 Source . . . . .                  | 51 |
| 3.59.2 Description . . . . .             | 51 |
| 3.59.3 Requirements Area . . . . .       | 51 |
| 3.59.4 Importance . . . . .              | 51 |
| 3.59.5 Applicable Standard . . . . .     | 51 |
| 3.59.6 Constraints . . . . .             | 51 |
| 3.60 Task Priority Selection . . . . .   | 51 |
| 3.60.1 Source . . . . .                  | 51 |
| 3.60.2 Description . . . . .             | 51 |
| 3.60.3 Requirements Area . . . . .       | 51 |
| 3.60.4 Importance . . . . .              | 51 |
| 3.60.5 Applicable Standard . . . . .     | 51 |
| 3.60.6 Constraints . . . . .             | 51 |
| 3.61 Task Status Selection . . . . .     | 51 |
| 3.61.1 Source . . . . .                  | 51 |
| 3.61.2 Description . . . . .             | 51 |
| 3.61.3 Requirements Area . . . . .       | 51 |
| 3.61.4 Importance . . . . .              | 52 |
| 3.61.5 Applicable Standard . . . . .     | 52 |
| 3.61.6 Constraints . . . . .             | 52 |
| 3.62 Task Start Date Selection . . . . . | 52 |
| 3.62.1 Source . . . . .                  | 52 |
| 3.62.2 Description . . . . .             | 52 |
| 3.62.3 Requirements Area . . . . .       | 52 |
| 3.62.4 Importance . . . . .              | 52 |
| 3.62.5 Applicable Standard . . . . .     | 52 |
| 3.62.6 Constraints . . . . .             | 52 |
| 3.63 Task Due Date Selection . . . . .   | 52 |
| 3.63.1 Source . . . . .                  | 52 |
| 3.63.2 Description . . . . .             | 52 |
| 3.63.3 Requirements Area . . . . .       | 52 |
| 3.63.4 Importance . . . . .              | 52 |
| 3.63.5 Applicable Standard . . . . .     | 52 |
| 3.63.6 Constraints . . . . .             | 52 |
| 3.64 Task Due Time Selection . . . . .   | 53 |
| 3.64.1 Source . . . . .                  | 53 |
| 3.64.2 Description . . . . .             | 53 |
| 3.64.3 Requirements Area . . . . .       | 53 |
| 3.64.4 Importance . . . . .              | 53 |
| 3.64.5 Applicable Standard . . . . .     | 53 |
| 3.64.6 Constraints . . . . .             | 53 |
| 3.65 New Task Menu Buttons . . . . .     | 53 |
| 3.65.1 Source . . . . .                  | 53 |
| 3.65.2 Description . . . . .             | 53 |
| 3.65.3 Requirements Area . . . . .       | 53 |
| 3.65.4 Importance . . . . .              | 53 |

|        |                                        |    |
|--------|----------------------------------------|----|
| 3.65.5 | Applicable Standard . . . . .          | 53 |
| 3.65.6 | Constraints . . . . .                  | 53 |
| 3.66   | Task Viewing and Editing . . . . .     | 53 |
| 3.66.1 | Source . . . . .                       | 53 |
| 3.66.2 | Description . . . . .                  | 53 |
| 3.66.3 | Requirements Area . . . . .            | 54 |
| 3.66.4 | Importance . . . . .                   | 54 |
| 3.66.5 | Applicable Standard . . . . .          | 54 |
| 3.66.6 | Constraints . . . . .                  | 54 |
| 3.67   | Task Display Format . . . . .          | 54 |
| 3.67.1 | Source . . . . .                       | 54 |
| 3.67.2 | Description . . . . .                  | 54 |
| 3.67.3 | Requirements Area . . . . .            | 54 |
| 3.67.4 | Importance . . . . .                   | 54 |
| 3.67.5 | Applicable Standard . . . . .          | 54 |
| 3.67.6 | Constraints . . . . .                  | 54 |
| 3.68   | Task Block Components . . . . .        | 54 |
| 3.68.1 | Source . . . . .                       | 54 |
| 3.68.2 | Description . . . . .                  | 54 |
| 3.68.3 | Requirements Area . . . . .            | 54 |
| 3.68.4 | Importance . . . . .                   | 54 |
| 3.68.5 | Applicable Standard . . . . .          | 54 |
| 3.68.6 | Constraints . . . . .                  | 55 |
| 3.69   | Task Editing Initiation . . . . .      | 55 |
| 3.69.1 | Source . . . . .                       | 55 |
| 3.69.2 | Description . . . . .                  | 55 |
| 3.69.3 | Requirements Area . . . . .            | 55 |
| 3.69.4 | Importance . . . . .                   | 55 |
| 3.69.5 | Applicable Standard . . . . .          | 55 |
| 3.69.6 | Constraints . . . . .                  | 55 |
| 3.70   | Edit Task Form Display . . . . .       | 55 |
| 3.70.1 | Source . . . . .                       | 55 |
| 3.70.2 | Description . . . . .                  | 55 |
| 3.70.3 | Requirements Area . . . . .            | 55 |
| 3.70.4 | Importance . . . . .                   | 55 |
| 3.70.5 | Applicable Standard . . . . .          | 55 |
| 3.70.6 | Constraints . . . . .                  | 55 |
| 3.71   | Edit Task Form Pre-fill . . . . .      | 55 |
| 3.71.1 | Source . . . . .                       | 55 |
| 3.71.2 | Description . . . . .                  | 56 |
| 3.71.3 | Requirements Area . . . . .            | 56 |
| 3.71.4 | Importance . . . . .                   | 56 |
| 3.71.5 | Applicable Standard . . . . .          | 56 |
| 3.71.6 | Constraints . . . . .                  | 56 |
| 3.72   | Edit Task Form Functionality . . . . . | 56 |
| 3.72.1 | Source . . . . .                       | 56 |
| 3.72.2 | Description . . . . .                  | 56 |
| 3.72.3 | Requirements Area . . . . .            | 56 |

|        |                                        |    |
|--------|----------------------------------------|----|
| 3.72.4 | Importance . . . . .                   | 56 |
| 3.72.5 | Applicable Standard . . . . .          | 56 |
| 3.72.6 | Constraints . . . . .                  | 56 |
| 3.73   | Edit Task Form Save Button . . . . .   | 56 |
| 3.73.1 | Source . . . . .                       | 56 |
| 3.73.2 | Description . . . . .                  | 56 |
| 3.73.3 | Requirements Area . . . . .            | 56 |
| 3.73.4 | Importance . . . . .                   | 56 |
| 3.73.5 | Applicable Standard . . . . .          | 57 |
| 3.73.6 | Constraints . . . . .                  | 57 |
| 3.74   | Edit Task Form Cancel Button . . . . . | 57 |
| 3.74.1 | Source . . . . .                       | 57 |
| 3.74.2 | Description . . . . .                  | 57 |
| 3.74.3 | Requirements Area . . . . .            | 57 |
| 3.74.4 | Importance . . . . .                   | 57 |
| 3.74.5 | Applicable Standard . . . . .          | 57 |
| 3.74.6 | Constraints . . . . .                  | 57 |
| 3.75   | Subject Page Title . . . . .           | 57 |
| 3.75.1 | Source . . . . .                       | 57 |
| 3.75.2 | Description . . . . .                  | 57 |
| 3.75.3 | Requirements Area . . . . .            | 57 |
| 3.75.4 | Importance . . . . .                   | 57 |
| 3.75.5 | Applicable Standard . . . . .          | 57 |
| 3.75.6 | Constraints . . . . .                  | 57 |
| 3.76   | View and Edit Subject . . . . .        | 57 |
| 3.76.1 | Source . . . . .                       | 57 |
| 3.76.2 | Description . . . . .                  | 58 |
| 3.76.3 | Requirements Area . . . . .            | 58 |
| 3.76.4 | Importance . . . . .                   | 58 |
| 3.76.5 | Applicable Standard . . . . .          | 58 |
| 3.76.6 | Constraints . . . . .                  | 58 |
| 3.77   | Creating Subjects . . . . .            | 58 |
| 3.77.1 | Source . . . . .                       | 58 |
| 3.77.2 | Description . . . . .                  | 58 |
| 3.77.3 | Requirements Area . . . . .            | 58 |
| 3.77.4 | Importance . . . . .                   | 58 |
| 3.77.5 | Applicable Standard . . . . .          | 58 |
| 3.77.6 | Constraints . . . . .                  | 58 |
| 3.78   | Subject List Format . . . . .          | 58 |
| 3.78.1 | Source . . . . .                       | 58 |
| 3.78.2 | Description . . . . .                  | 58 |
| 3.78.3 | Requirements Area . . . . .            | 58 |
| 3.78.4 | Importance . . . . .                   | 58 |
| 3.78.5 | Applicable Standard . . . . .          | 58 |
| 3.78.6 | Constraints . . . . .                  | 59 |
| 3.79   | Subject Edit Menu . . . . .            | 59 |
| 3.79.1 | Source . . . . .                       | 59 |
| 3.79.2 | Description . . . . .                  | 59 |

|                                                                      |    |
|----------------------------------------------------------------------|----|
| 3.79.3 Requirements Area . . . . .                                   | 59 |
| 3.79.4 Importance . . . . .                                          | 59 |
| 3.79.5 Applicable Standard . . . . .                                 | 59 |
| 3.79.6 Constraints . . . . .                                         | 59 |
| 3.80 Subject Deletion . . . . .                                      | 59 |
| 3.80.1 Source . . . . .                                              | 59 |
| 3.80.2 Description . . . . .                                         | 59 |
| 3.80.3 Requirements Area . . . . .                                   | 59 |
| 3.80.4 Importance . . . . .                                          | 59 |
| 3.80.5 Applicable Standard . . . . .                                 | 59 |
| 3.80.6 Constraints . . . . .                                         | 59 |
| 3.81 Subject Archiving . . . . .                                     | 59 |
| 3.81.1 Source . . . . .                                              | 59 |
| 3.81.2 Description . . . . .                                         | 59 |
| 3.81.3 Requirements Area . . . . .                                   | 60 |
| 3.81.4 Importance . . . . .                                          | 60 |
| 3.81.5 Applicable Standard . . . . .                                 | 60 |
| 3.81.6 Constraints . . . . .                                         | 60 |
| 3.82 Subject Editing . . . . .                                       | 60 |
| 3.82.1 Source . . . . .                                              | 60 |
| 3.82.2 Description . . . . .                                         | 60 |
| 3.82.3 Requirements Area . . . . .                                   | 60 |
| 3.82.4 Importance . . . . .                                          | 60 |
| 3.82.5 Applicable Standard . . . . .                                 | 60 |
| 3.82.6 Constraints . . . . .                                         | 60 |
| 3.83 Saving New Subjects . . . . .                                   | 60 |
| 3.83.1 Source . . . . .                                              | 60 |
| 3.83.2 Description . . . . .                                         | 60 |
| 3.83.3 Requirements Area . . . . .                                   | 60 |
| 3.83.4 Importance . . . . .                                          | 60 |
| 3.83.5 Applicable Standard . . . . .                                 | 60 |
| 3.83.6 Constraints . . . . .                                         | 61 |
| 3.84 Subject Creation Canceling . . . . .                            | 61 |
| 3.84.1 Source . . . . .                                              | 61 |
| 3.84.2 Description . . . . .                                         | 61 |
| 3.84.3 Requirements Area . . . . .                                   | 61 |
| 3.84.4 Importance . . . . .                                          | 61 |
| 3.84.5 Applicable Standard . . . . .                                 | 61 |
| 3.84.6 Constraints . . . . .                                         | 61 |
| 3.85 Subject Filtering . . . . .                                     | 61 |
| 3.85.1 Source . . . . .                                              | 61 |
| 3.85.2 Description . . . . .                                         | 61 |
| 3.85.3 Requirements Area . . . . .                                   | 61 |
| 3.85.4 Importance . . . . .                                          | 61 |
| 3.85.5 Applicable Standard . . . . .                                 | 61 |
| 3.85.6 Constraints . . . . .                                         | 61 |
| 3.86 Filter Subjects by Subject, Semester, and Description . . . . . | 61 |
| 3.86.1 Source . . . . .                                              | 61 |

|        |                                        |    |
|--------|----------------------------------------|----|
| 3.86.2 | Description . . . . .                  | 62 |
| 3.86.3 | Requirements Area . . . . .            | 62 |
| 3.86.4 | Importance . . . . .                   | 62 |
| 3.86.5 | Applicable Standard . . . . .          | 62 |
| 3.86.6 | Constraints . . . . .                  | 62 |
| 3.87   | Clear Subject Filters . . . . .        | 62 |
| 3.87.1 | Source . . . . .                       | 62 |
| 3.87.2 | Description . . . . .                  | 62 |
| 3.87.3 | Requirements Area . . . . .            | 62 |
| 3.87.4 | Importance . . . . .                   | 62 |
| 3.87.5 | Applicable Standard . . . . .          | 62 |
| 3.87.6 | Constraints . . . . .                  | 62 |
| 3.88   | Subject Creation . . . . .             | 62 |
| 3.88.1 | Source . . . . .                       | 62 |
| 3.88.2 | Description . . . . .                  | 62 |
| 3.88.3 | Requirements Area . . . . .            | 62 |
| 3.88.4 | Importance . . . . .                   | 62 |
| 3.88.5 | Applicable Standard . . . . .          | 63 |
| 3.88.6 | Constraints . . . . .                  | 63 |
| 3.89   | Creation Menu Uniformity . . . . .     | 63 |
| 3.89.1 | Source . . . . .                       | 63 |
| 3.89.2 | Description . . . . .                  | 63 |
| 3.89.3 | Requirements Area . . . . .            | 63 |
| 3.89.4 | Importance . . . . .                   | 63 |
| 3.89.5 | Applicable Standard . . . . .          | 63 |
| 3.89.6 | Constraints . . . . .                  | 63 |
| 3.90   | Internal Subject Links . . . . .       | 63 |
| 3.90.1 | Source . . . . .                       | 63 |
| 3.90.2 | Description . . . . .                  | 63 |
| 3.90.3 | Requirements Area . . . . .            | 63 |
| 3.90.4 | Importance . . . . .                   | 63 |
| 3.90.5 | Applicable Standard . . . . .          | 63 |
| 3.90.6 | Constraints . . . . .                  | 63 |
| 3.91   | Edit Task Form Display . . . . .       | 64 |
| 3.91.1 | Source . . . . .                       | 64 |
| 3.91.2 | Description . . . . .                  | 64 |
| 3.91.3 | Requirements Area . . . . .            | 64 |
| 3.91.4 | Importance . . . . .                   | 64 |
| 3.91.5 | Applicable Standard . . . . .          | 64 |
| 3.91.6 | Constraints . . . . .                  | 64 |
| 3.92   | Pre-fill Edit Task Form . . . . .      | 64 |
| 3.92.1 | Source . . . . .                       | 64 |
| 3.92.2 | Description . . . . .                  | 64 |
| 3.92.3 | Requirements Area . . . . .            | 64 |
| 3.92.4 | Importance . . . . .                   | 64 |
| 3.92.5 | Applicable Standard . . . . .          | 64 |
| 3.92.6 | Constraints . . . . .                  | 64 |
| 3.93   | Edit Task Form Functionality . . . . . | 64 |

|         |                                    |    |
|---------|------------------------------------|----|
| 3.93.1  | Source                             | 64 |
| 3.93.2  | Description                        | 64 |
| 3.93.3  | Requirements Area                  | 64 |
| 3.93.4  | Importance                         | 64 |
| 3.93.5  | Applicable Standard                | 65 |
| 3.93.6  | Constraints                        | 65 |
| 3.94    | Save Updated Task                  | 65 |
| 3.94.1  | Source                             | 65 |
| 3.94.2  | Description                        | 65 |
| 3.94.3  | Requirements Area                  | 65 |
| 3.94.4  | Importance                         | 65 |
| 3.94.5  | Applicable Standard                | 65 |
| 3.94.6  | Constraints                        | 65 |
| 3.95    | Cancel Edit Task                   | 65 |
| 3.95.1  | Source                             | 65 |
| 3.95.2  | Description                        | 65 |
| 3.95.3  | Requirements Area                  | 65 |
| 3.95.4  | Importance                         | 65 |
| 3.95.5  | Applicable Standard                | 65 |
| 3.95.6  | Constraints                        | 65 |
| 3.96    | Calendar Color Schema Requirements | 65 |
| 3.96.1  | Priority                           | 65 |
| 3.96.2  | Description                        | 66 |
| 3.96.3  | Source                             | 66 |
| 3.96.4  | Applicable Standards               | 66 |
| 3.96.5  | Constraints                        | 66 |
| 3.97    | Calendar with Tasks                | 66 |
| 3.97.1  | Priority                           | 66 |
| 3.97.2  | Description                        | 66 |
| 3.97.3  | Source                             | 66 |
| 3.97.4  | Applicable Standards               | 66 |
| 3.97.5  | Constraints                        | 66 |
| 3.98    | Task Display on Calendar           | 66 |
| 3.98.1  | Priority                           | 66 |
| 3.98.2  | Description                        | 66 |
| 3.98.3  | Source                             | 66 |
| 3.98.4  | Applicable Standards               | 66 |
| 3.98.5  | Constraints                        | 66 |
| 3.99    | Calendar Date Range                | 67 |
| 3.99.1  | Priority                           | 67 |
| 3.99.2  | Description                        | 67 |
| 3.99.3  | Source                             | 67 |
| 3.99.4  | Applicable Standards               | 67 |
| 3.99.5  | Constraints                        | 67 |
| 3.100   | Calendar View Options              | 67 |
| 3.100.1 | Priority                           | 67 |
| 3.100.2 | Description                        | 67 |
| 3.100.3 | Source                             | 67 |

|                                                         |    |
|---------------------------------------------------------|----|
| 3.100.4Applicable Standards . . . . .                   | 67 |
| 3.100.5Constraints . . . . .                            | 67 |
| 3.101Calendar Navigation . . . . .                      | 67 |
| 3.101.1Priority . . . . .                               | 67 |
| 3.101.2Description . . . . .                            | 68 |
| 3.101.3Source . . . . .                                 | 68 |
| 3.101.4Applicable Standards . . . . .                   | 68 |
| 3.101.5Constraints . . . . .                            | 68 |
| 3.102Add Task on Click of Day . . . . .                 | 68 |
| 3.102.1Priority . . . . .                               | 68 |
| 3.102.2Description . . . . .                            | 68 |
| 3.102.3Source . . . . .                                 | 68 |
| 3.102.4Applicable Standards . . . . .                   | 68 |
| 3.102.5Constraints . . . . .                            | 68 |
| 3.103Add Task from Calendar . . . . .                   | 68 |
| 3.103.1Priority . . . . .                               | 68 |
| 3.103.2Description . . . . .                            | 68 |
| 3.103.3Source . . . . .                                 | 68 |
| 3.103.4Applicable Standards . . . . .                   | 69 |
| 3.103.5Constraints . . . . .                            | 69 |
| 3.104Create Task from Input . . . . .                   | 69 |
| 3.104.1Priority . . . . .                               | 69 |
| 3.104.2Description . . . . .                            | 69 |
| 3.104.3Source . . . . .                                 | 69 |
| 3.104.4Applicable Standards . . . . .                   | 69 |
| 3.104.5Constraints . . . . .                            | 69 |
| 3.105Calendar Initialization . . . . .                  | 69 |
| 3.105.1Priority . . . . .                               | 69 |
| 3.105.2Description . . . . .                            | 69 |
| 3.105.3Source . . . . .                                 | 69 |
| 3.105.4Applicable Standards . . . . .                   | 69 |
| 3.105.5Constraints . . . . .                            | 69 |
| 3.106Add Subject from Add Task Form . . . . .           | 69 |
| 3.106.1Priority . . . . .                               | 69 |
| 3.106.2Description . . . . .                            | 69 |
| 3.106.3Source . . . . .                                 | 70 |
| 3.106.4Applicable Standards . . . . .                   | 70 |
| 3.106.5Constraints . . . . .                            | 70 |
| 3.107Add Subject from Add Task Form . . . . .           | 70 |
| 3.107.1Priority . . . . .                               | 70 |
| 3.107.2Description . . . . .                            | 70 |
| 3.107.3Source . . . . .                                 | 70 |
| 3.107.4Applicable Standards . . . . .                   | 70 |
| 3.107.5Constraints . . . . .                            | 70 |
| 3.108Edit and Delete Tasks from Calendar View . . . . . | 70 |
| 3.108.1Priority . . . . .                               | 70 |
| 3.108.2Description . . . . .                            | 70 |
| 3.108.3Source . . . . .                                 | 70 |

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| 3.108.4  | Applicable Standards . . . . .                                             | 70        |
| 3.108.5  | Constraints . . . . .                                                      | 70        |
| 3.109    | Calendar Format of Current Day (Mobile) . . . . .                          | 71        |
| 3.109.1  | Priority . . . . .                                                         | 71        |
| 3.109.2  | Description . . . . .                                                      | 71        |
| 3.109.3  | Source . . . . .                                                           | 71        |
| 3.109.4  | Applicable Standards . . . . .                                             | 71        |
| 3.109.5  | Constraints . . . . .                                                      | 71        |
| 3.110    | Calendar Format of Out-of-Range Dates (Mobile) . . . . .                   | 71        |
| 3.110.1  | Priority . . . . .                                                         | 71        |
| 3.110.2  | Description . . . . .                                                      | 71        |
| 3.110.3  | Source . . . . .                                                           | 71        |
| 3.110.4  | Applicable Standards . . . . .                                             | 71        |
| 3.110.5  | Constraints . . . . .                                                      | 71        |
| <b>4</b> | <b>Packaging Requirements</b>                                              | <b>72</b> |
| 4.1      | Progressive Web Application (PWA) Delivery . . . . .                       | 72        |
| 4.1.1    | Description . . . . .                                                      | 72        |
| 4.1.2    | Source . . . . .                                                           | 72        |
| 4.1.3    | Constraints . . . . .                                                      | 72        |
| 4.1.4    | Standards . . . . .                                                        | 72        |
| 4.1.5    | Priority . . . . .                                                         | 72        |
| 4.2      | Direct Download and Access . . . . .                                       | 72        |
| 4.2.1    | Description . . . . .                                                      | 72        |
| 4.2.2    | Source . . . . .                                                           | 72        |
| 4.2.3    | Constraints . . . . .                                                      | 72        |
| 4.2.4    | Standards . . . . .                                                        | 72        |
| 4.2.5    | Priority . . . . .                                                         | 72        |
| 4.3      | Firebase Pre-Configured . . . . .                                          | 72        |
| 4.3.1    | Description . . . . .                                                      | 73        |
| 4.3.2    | Source . . . . .                                                           | 73        |
| 4.3.3    | Constraints . . . . .                                                      | 73        |
| 4.3.4    | Standards . . . . .                                                        | 73        |
| 4.3.5    | Priority . . . . .                                                         | 73        |
| 4.4      | No Third-Party Software for Installation or Use . . . . .                  | 73        |
| 4.4.1    | Description . . . . .                                                      | 73        |
| 4.4.2    | Source . . . . .                                                           | 73        |
| 4.4.3    | Constraints . . . . .                                                      | 73        |
| 4.4.4    | Standards . . . . .                                                        | 73        |
| 4.4.5    | Priority . . . . .                                                         | 73        |
| 4.5      | APK Package (Android) Delivery . . . . .                                   | 73        |
| 4.5.1    | Description . . . . .                                                      | 73        |
| 4.5.2    | Source . . . . .                                                           | 73        |
| 4.5.3    | Constraints . . . . .                                                      | 73        |
| 4.5.4    | Standards . . . . .                                                        | 73        |
| 4.5.5    | Priority . . . . .                                                         | 74        |
| 4.6      | Mobile Version to Be Distributed via Google Play Store (Android) . . . . . | 74        |
| 4.6.1    | Description . . . . .                                                      | 74        |

|          |                                             |           |
|----------|---------------------------------------------|-----------|
| 4.6.2    | Source . . . . .                            | 74        |
| 4.6.3    | Constraints . . . . .                       | 74        |
| 4.6.4    | Standards . . . . .                         | 74        |
| 4.6.5    | Priority . . . . .                          | 74        |
| <b>5</b> | <b>Performance Requirements</b>             | <b>75</b> |
| 5.1      | Optimized Loading Time . . . . .            | 75        |
| 5.1.1    | Description . . . . .                       | 75        |
| 5.1.2    | Source . . . . .                            | 75        |
| 5.1.3    | Constraints . . . . .                       | 75        |
| 5.1.4    | Standards . . . . .                         | 75        |
| 5.1.5    | Priority . . . . .                          | 75        |
| 5.2      | Fast Loading on Slow Networks . . . . .     | 75        |
| 5.2.1    | Description . . . . .                       | 75        |
| 5.2.2    | Source . . . . .                            | 75        |
| 5.2.3    | Constraints . . . . .                       | 75        |
| 5.2.4    | Standards . . . . .                         | 75        |
| 5.2.5    | Priority . . . . .                          | 75        |
| 5.3      | Efficient Memory Usage . . . . .            | 75        |
| 5.3.1    | Description . . . . .                       | 75        |
| 5.3.2    | Source . . . . .                            | 75        |
| 5.3.3    | Constraints . . . . .                       | 76        |
| 5.3.4    | Standards . . . . .                         | 76        |
| 5.3.5    | Priority . . . . .                          | 76        |
| 5.4      | Low Latency for Real-Time Updates . . . . . | 76        |
| 5.4.1    | Description . . . . .                       | 76        |
| 5.4.2    | Source . . . . .                            | 76        |
| 5.4.3    | Constraints . . . . .                       | 76        |
| 5.4.4    | Standards . . . . .                         | 76        |
| 5.4.5    | Priority . . . . .                          | 76        |
| 5.5      | Background Data Sync . . . . .              | 76        |
| 5.5.1    | Description . . . . .                       | 76        |
| 5.5.2    | Source . . . . .                            | 76        |
| 5.5.3    | Constraints . . . . .                       | 76        |
| 5.5.4    | Standards . . . . .                         | 76        |
| 5.5.5    | Priority . . . . .                          | 76        |
| 5.6      | Offline Mode . . . . .                      | 76        |
| 5.6.1    | Source . . . . .                            | 76        |
| 5.6.2    | Description . . . . .                       | 76        |
| 5.6.3    | Requirements Area . . . . .                 | 77        |
| 5.6.4    | Importance . . . . .                        | 77        |
| 5.6.5    | Applicable Standard . . . . .               | 77        |
| 5.6.6    | Constraints . . . . .                       | 77        |
| 5.7      | Background Sync . . . . .                   | 77        |
| 5.7.1    | Source . . . . .                            | 77        |
| 5.7.2    | Description . . . . .                       | 77        |
| 5.7.3    | Requirements Area . . . . .                 | 77        |
| 5.7.4    | Importance . . . . .                        | 77        |

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| 5.7.5    | Applicable Standard . . . . .                  | 77        |
| 5.7.6    | Constraints . . . . .                          | 77        |
| 5.8      | Fetch Archived Data . . . . .                  | 77        |
| 5.8.1    | Source . . . . .                               | 77        |
| 5.8.2    | Description . . . . .                          | 77        |
| 5.8.3    | Requirements Area . . . . .                    | 77        |
| 5.8.4    | Importance . . . . .                           | 77        |
| 5.8.5    | Applicable Standard . . . . .                  | 77        |
| 5.8.6    | Constraints . . . . .                          | 77        |
| 5.9      | Task Creation in Firebase . . . . .            | 78        |
| 5.9.1    | Source . . . . .                               | 78        |
| 5.9.2    | Description . . . . .                          | 78        |
| 5.9.3    | Requirements Area . . . . .                    | 78        |
| 5.9.4    | Importance . . . . .                           | 78        |
| 5.9.5    | Applicable Standard . . . . .                  | 78        |
| 5.9.6    | Constraints . . . . .                          | 78        |
| 5.10     | Subject Creation in Firebase . . . . .         | 78        |
| 5.10.1   | Source . . . . .                               | 78        |
| 5.10.2   | Description . . . . .                          | 78        |
| 5.10.3   | Requirements Area . . . . .                    | 78        |
| 5.10.4   | Importance . . . . .                           | 78        |
| 5.10.5   | Applicable Standard . . . . .                  | 78        |
| 5.10.6   | Constraints . . . . .                          | 78        |
| 5.11     | Project Creation in Firebase . . . . .         | 78        |
| 5.11.1   | Source . . . . .                               | 78        |
| 5.11.2   | Description . . . . .                          | 79        |
| 5.11.3   | Requirements Area . . . . .                    | 79        |
| 5.11.4   | Importance . . . . .                           | 79        |
| 5.11.5   | Applicable Standard . . . . .                  | 79        |
| 5.11.6   | Constraints . . . . .                          | 79        |
| <b>6</b> | <b>Safety Requirements</b>                     | <b>80</b> |
| <b>7</b> | <b>Security Requirements</b>                   | <b>81</b> |
| 7.1      | HTTPS Enforcement . . . . .                    | 81        |
| 7.1.1    | Description . . . . .                          | 81        |
| 7.1.2    | Source . . . . .                               | 81        |
| 7.1.3    | Constraints . . . . .                          | 81        |
| 7.1.4    | Standards . . . . .                            | 81        |
| 7.1.5    | Priority . . . . .                             | 81        |
| 7.2      | Email Notification Security . . . . .          | 81        |
| 7.2.1    | Description . . . . .                          | 81        |
| 7.2.2    | Source . . . . .                               | 81        |
| 7.2.3    | Constraints . . . . .                          | 81        |
| 7.2.4    | Standards . . . . .                            | 81        |
| 7.2.5    | Priority . . . . .                             | 81        |
| 7.3      | Compliance with Bulk Email Standards . . . . . | 81        |
| 7.3.1    | Description . . . . .                          | 81        |

|        |                                       |    |
|--------|---------------------------------------|----|
| 7.3.2  | Source . . . . .                      | 81 |
| 7.3.3  | Constraints . . . . .                 | 82 |
| 7.3.4  | Standards . . . . .                   | 82 |
| 7.3.5  | Priority . . . . .                    | 82 |
| 7.4    | Secure Login Form . . . . .           | 82 |
| 7.4.1  | Source . . . . .                      | 82 |
| 7.4.2  | Description . . . . .                 | 82 |
| 7.4.3  | Requirements Area . . . . .           | 82 |
| 7.4.4  | Importance . . . . .                  | 82 |
| 7.4.5  | Applicable Standard . . . . .         | 82 |
| 7.4.6  | Constraints . . . . .                 | 82 |
| 7.5    | Password Masking . . . . .            | 82 |
| 7.5.1  | Source . . . . .                      | 82 |
| 7.5.2  | Description . . . . .                 | 82 |
| 7.5.3  | Requirements Area . . . . .           | 82 |
| 7.5.4  | Importance . . . . .                  | 82 |
| 7.5.5  | Applicable Standard . . . . .         | 82 |
| 7.5.6  | Constraints . . . . .                 | 82 |
| 7.6    | Password Reset Option . . . . .       | 83 |
| 7.6.1  | Source . . . . .                      | 83 |
| 7.6.2  | Description . . . . .                 | 83 |
| 7.6.3  | Requirements Area . . . . .           | 83 |
| 7.6.4  | Importance . . . . .                  | 83 |
| 7.6.5  | Applicable Standard . . . . .         | 83 |
| 7.6.6  | Constraints . . . . .                 | 83 |
| 7.7    | Brute Force Protection . . . . .      | 83 |
| 7.7.1  | Source . . . . .                      | 83 |
| 7.7.2  | Description . . . . .                 | 83 |
| 7.7.3  | Requirements Area . . . . .           | 83 |
| 7.7.4  | Importance . . . . .                  | 83 |
| 7.7.5  | Applicable Standard . . . . .         | 83 |
| 7.7.6  | Constraints . . . . .                 | 83 |
| 7.8    | Login Activity Logging . . . . .      | 83 |
| 7.8.1  | Source . . . . .                      | 83 |
| 7.8.2  | Description . . . . .                 | 83 |
| 7.8.3  | Requirements Area . . . . .           | 83 |
| 7.8.4  | Importance . . . . .                  | 83 |
| 7.8.5  | Applicable Standard . . . . .         | 84 |
| 7.8.6  | Constraints . . . . .                 | 84 |
| 7.9    | Password Strength Indicator . . . . . | 84 |
| 7.9.1  | Source . . . . .                      | 84 |
| 7.9.2  | Description . . . . .                 | 84 |
| 7.9.3  | Requirements Area . . . . .           | 84 |
| 7.9.4  | Importance . . . . .                  | 84 |
| 7.9.5  | Applicable Standard . . . . .         | 84 |
| 7.9.6  | Constraints . . . . .                 | 84 |
| 7.10   | Email Verification . . . . .          | 84 |
| 7.10.1 | Source . . . . .                      | 84 |

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| 7.10.2   | Description . . . . .                         | 84        |
| 7.10.3   | Requirements Area . . . . .                   | 84        |
| 7.10.4   | Importance . . . . .                          | 84        |
| 7.10.5   | Applicable Standard . . . . .                 | 84        |
| 7.10.6   | Constraints . . . . .                         | 84        |
| 7.11     | CAPTCHA Implementation . . . . .              | 84        |
| 7.11.1   | Source . . . . .                              | 84        |
| 7.11.2   | Description . . . . .                         | 84        |
| 7.11.3   | Requirements Area . . . . .                   | 85        |
| 7.11.4   | Importance . . . . .                          | 85        |
| 7.11.5   | Applicable Standard . . . . .                 | 85        |
| 7.11.6   | Constraints . . . . .                         | 85        |
| 7.12     | User Authentication and Logout . . . . .      | 85        |
| 7.12.1   | Source . . . . .                              | 85        |
| 7.12.2   | Description . . . . .                         | 85        |
| 7.12.3   | Requirements Area . . . . .                   | 85        |
| 7.12.4   | Importance . . . . .                          | 85        |
| 7.12.5   | Applicable Standard . . . . .                 | 85        |
| 7.12.6   | Constraints . . . . .                         | 85        |
| <b>8</b> | <b>Maintenance &amp; Support Requirements</b> | <b>86</b> |
| 8.1      | Performance Auditing Tools . . . . .          | 86        |
| 8.1.1    | Description . . . . .                         | 86        |
| 8.1.2    | Source . . . . .                              | 86        |
| 8.1.3    | Constraints . . . . .                         | 86        |
| 8.1.4    | Standards . . . . .                           | 86        |
| 8.1.5    | Priority . . . . .                            | 86        |
| 8.2      | Version Control Support . . . . .             | 86        |
| 8.2.1    | Description . . . . .                         | 86        |
| 8.2.2    | Source . . . . .                              | 86        |
| 8.2.3    | Constraints . . . . .                         | 86        |
| 8.2.4    | Standards . . . . .                           | 86        |
| 8.2.5    | Priority . . . . .                            | 86        |
| 8.3      | Notification Storage . . . . .                | 87        |
| 8.3.1    | Description . . . . .                         | 87        |
| 8.3.2    | Source . . . . .                              | 87        |
| 8.3.3    | Constraints . . . . .                         | 87        |
| 8.3.4    | Standards . . . . .                           | 87        |
| 8.3.5    | Priority . . . . .                            | 87        |
| <b>9</b> | <b>Other Requirements</b>                     | <b>88</b> |
| 9.1      | PWA Manifest File . . . . .                   | 88        |
| 9.1.1    | Description . . . . .                         | 88        |
| 9.1.2    | Source . . . . .                              | 88        |
| 9.1.3    | Constraints . . . . .                         | 88        |
| 9.1.4    | Standards . . . . .                           | 88        |
| 9.1.5    | Priority . . . . .                            | 88        |

|                                               |           |
|-----------------------------------------------|-----------|
| <b>10 Future Items</b>                        | <b>89</b> |
| 10.1 Background Data Sync . . . . .           | 89        |
| 10.1.1 Description . . . . .                  | 89        |
| 10.1.2 Source . . . . .                       | 89        |
| 10.1.3 Constraints . . . . .                  | 89        |
| 10.1.4 Standards . . . . .                    | 89        |
| 10.1.5 Priority . . . . .                     | 89        |
| 10.2 Social Media Login Integration . . . . . | 89        |
| 10.2.1 Source . . . . .                       | 89        |
| 10.2.2 Description . . . . .                  | 89        |
| 10.2.3 Requirements Area . . . . .            | 89        |
| 10.2.4 Importance . . . . .                   | 89        |
| 10.2.5 Applicable Standard . . . . .          | 89        |
| 10.2.6 Constraints . . . . .                  | 89        |
| 10.3 Widget Support . . . . .                 | 89        |
| 10.3.1 Source . . . . .                       | 89        |
| 10.3.2 Description . . . . .                  | 89        |
| 10.3.3 Requirements Area . . . . .            | 90        |
| 10.3.4 Importance . . . . .                   | 90        |
| 10.3.5 Applicable Standard . . . . .          | 90        |
| 10.3.6 Constraints . . . . .                  | 90        |
| 10.4 Voice Input for Tasks . . . . .          | 90        |
| 10.4.1 Source . . . . .                       | 90        |
| 10.4.2 Description . . . . .                  | 90        |
| 10.4.3 Requirements Area . . . . .            | 90        |
| 10.4.4 Importance . . . . .                   | 90        |
| 10.4.5 Applicable Standard . . . . .          | 90        |
| 10.4.6 Constraints . . . . .                  | 90        |

## LIST OF FIGURES

|   |                                           |    |
|---|-------------------------------------------|----|
| 1 | Task Page Conceptual Drawing . . . . .    | 25 |
| 2 | Project Page Conceptual Drawing . . . . . | 26 |
| 3 | Subject Page Conceptual Drawing . . . . . | 26 |

# 1 PRODUCT CONCEPT

The Learnleaf Organizer is a system designed to streamline academic management for students, teaching assistants, and educators. Students can use the tool to organize and track assignments and project deadlines, helping them stay on top of their academic responsibilities. [1]

## 1.1 PURPOSE AND USE

Students will use it to input their assignment deadlines and project timelines, with the ability to set reminders and receive notifications. This feature helps them break down their workload, prioritize tasks, and stay on top of their academic commitments. The tool's dashboard provides a clear overview of upcoming deadlines and tasks, helping users plan ahead and avoid last-minute cramming. Overall, the Learnleaf Organizer acts as a centralized academic management platform, enabling users to stay organized, productive, and up-to-date with their academic responsibilities. [1]

## 1.2 INTENDED AUDIENCE

The primary audience of the Learnleaf Organizer is students, but this can be extended to teaching assistants and teachers as well. They will benefit by being able to keep tabs on grading timelines, and track class progress. Through the Learnleaf Organizer, users will be able to enhance their time management, productivity, and overall academic performance. [1]

The image shows a conceptual drawing of a task management interface. At the top, there is a 'Filters' dropdown menu. Below it, there is a grid of task cards. Each card has a title, a subject dropdown, a project dropdown, priority and status dropdowns, start and end date pickers, a time due input, and 'SAVE' and 'DELETE' buttons. The cards are titled: 'PDAs and Turing Machines', 'Sprint Plan #3', 'Homework 6', 'UC3: Analyze Booth Lease Status', 'TM Variants', 'Compilers Exam 2', 'UC3: Team Member Evaluation', and 'Homework 7'. A blue circular button with a plus sign is located at the bottom right of the grid.

| Task Title                      | Subject  | Project          | Priority | Status      | Start Date | Deadline   | Time Due | Actions      |
|---------------------------------|----------|------------------|----------|-------------|------------|------------|----------|--------------|
| PDAs and Turing Machines        | CSE-3315 | ToC Section 002  | Medium   | Not Started | 14/10/2024 | 18/10/2024 | 09:20    | SAVE, DELETE |
| Sprint Plan #3                  | CSE-4316 | Senior Design    | Medium   | Not Started | 14/10/2024 | 21/10/2024 | 10:50    | SAVE, DELETE |
| Homework 6                      | CSE-4321 |                  | Medium   | Not Started | 17/10/2024 | 24/10/2024 | --:--    | SAVE, DELETE |
| UC3: Analyze Booth Lease Status | CSE-4361 | CSE-4361 Project | Medium   | Not Started | dd/mm/yyyy | 24/10/2024 | 23:59    | SAVE, DELETE |
| TM Variants                     |          |                  |          |             |            |            |          |              |
| Compilers Exam 2                |          |                  |          |             |            |            |          |              |
| UC3: Team Member Evaluation     |          |                  |          |             |            |            |          |              |
| Homework 7                      |          |                  |          |             |            |            |          |              |

Figure 1: Task Page Conceptual Drawing

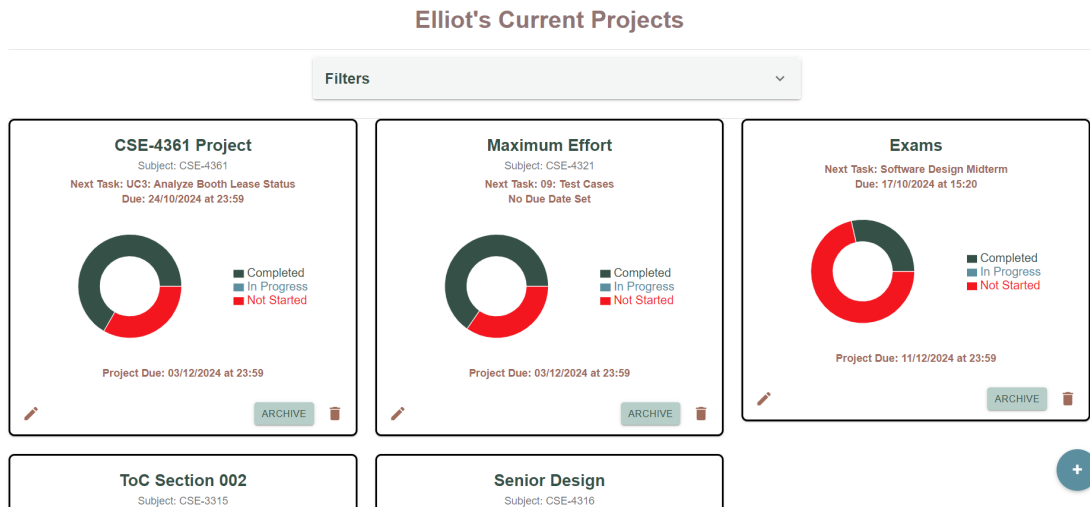


Figure 2: Project Page Conceptual Drawing

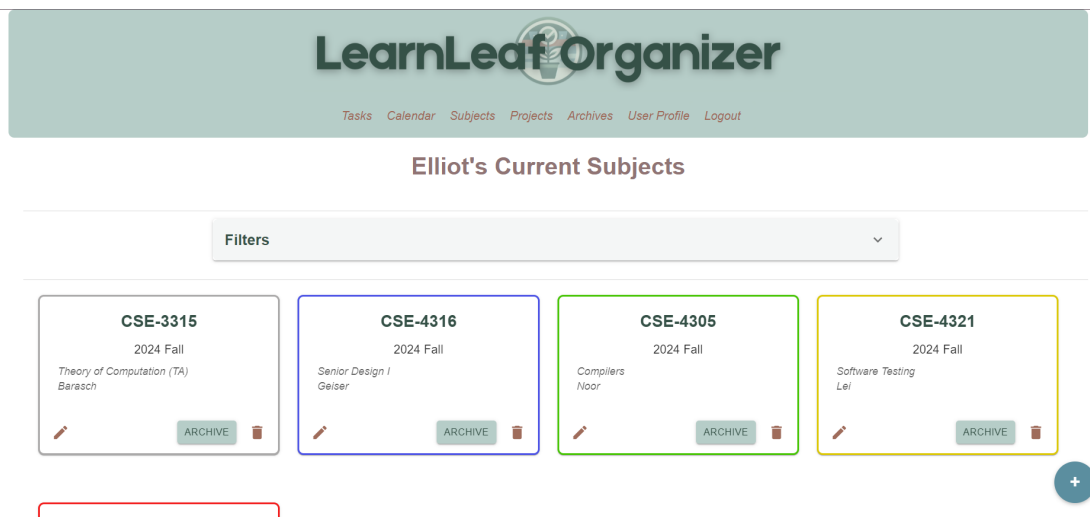


Figure 3: Subject Page Conceptual Drawing

## 2 PRODUCT DESCRIPTION

The LearnLeaf Organizer is an academic tool designed to help students efficiently manage projects, tasks, and assignments. It is built with a focus on organization, collaboration, and progress tracking, while ensuring a seamless user experience across multiple devices.

### 2.1 FEATURES & FUNCTIONS

The LearnLeaf Organizer provides a wide range of features aimed at improving productivity and keeping users on top of their academic responsibilities:

- **Project Management:** Enables users to create, edit, delete, and organize projects. Each project can contain multiple tasks, and users can dynamically reorder them as needed.
- **Task Tracking:** Users can track the progress of tasks within each project, with visual indicators showing task status, deadlines, and completion rates.

- **Notification System:** Sends push notifications to remind users of upcoming deadlines and task status updates. Users can set daily, weekly, and urgent reminders.
- **Mobile-First Design:** The platform is responsive and optimized for mobile devices, ensuring usability on smartphones and tablets.
- **Authentication:** Ensures secure access to the platform.
- **Data Synchronization:** The organizer synchronizes data across devices, ensuring users have access to updated information from anywhere.
- **Progressive Web App (PWA):** Supports offline functionality, allowing users to access and manage their tasks even without an active internet connection. It can also be installed on home screens for easier access.

These features are designed to help students and educators stay organized, improve task management, and streamline communication and progress tracking.

2.2 EXTERNAL INPUTS & OUTPUTS

The LearnLeaf Organizer relies on several external inputs and outputs to deliver its functionality:

Inputs:

- User inputs such as creating projects, managing tasks, and setting notification preferences.
- Authentication inputs through biometric sensors or manual login credentials.
- Synchronization data from other devices for ensuring a consistent experience across all platforms.

Outputs:

- Notifications sent to users regarding task deadlines, reminders, and updates.
- Display of real-time project progress and task completion indicators on the dashboard.
- Data outputs such as filtered project views and progress summaries.

| Name                | Description                                  | Use                               |
|---------------------|----------------------------------------------|-----------------------------------|
| User Projects       | Stores academic project information          | Input for project management      |
| Task Details        | Data on tasks including deadlines and status | Input for task management         |
| Notifications       | Push notifications for reminders             | Output to notify users            |
| Authentication Data | Biometric or login credentials               | Input for secure authentication   |
| Sync Data           | Synchronized data across devices             | Input/Output for multi-device use |

Table 2: Critical External Inputs and Outputs

2.3 PRODUCT INTERFACES

The LearnLeaf Organizer features intuitive and user-friendly interfaces to make managing academic tasks and projects easy for both students and educators:

- **Dashboard:** A central hub displaying active projects, upcoming deadlines, and progress tracking. Users can filter tasks by title, priority, status, start date, or due date.

- **Project and Task Management:** An interface that allows users to create, modify, and delete projects or tasks. The user-friendly design simplifies adding task details, setting deadlines, and updating task status.
- **Notification Settings:** Users can configure their notification preferences, determining how and when they are reminded of upcoming tasks or deadlines.
- **Mobile Interface:** The organizer is optimized for mobile devices, providing touch-friendly buttons, panels, and interactions that ensure ease of use on the go.

## 3 CUSTOMER REQUIREMENTS

Include a header paragraph specific to your product here. Customer requirements are those required features and functions specified for and by the intended audience for this product. This section establishes, clearly and concisely, the "look and feel" of the product, what each potential end-user should expect the product do and/or not do. Each requirement specified in this section is associated with a specific customer need that will be satisfied. In general Customer Requirements are the directly observable features and functions of the product that will be encountered by its users. Requirements specified in this section are created with, and must not be changed without, specific agreement of the intended customer/user/sponsor.

### 3.1 SERVICE WORKER FOR OFFLINE SUPPORT

#### 3.1.1 DESCRIPTION

The system shall use a service worker to cache essential assets and allow users to interact with the PWA even when offline.

#### 3.1.2 SOURCE

Customer Meeting

#### 3.1.3 CONSTRAINTS

The service worker must handle dynamic content efficiently and update cached content periodically.

#### 3.1.4 STANDARDS

W3C Service Worker API, PWA best practices

#### 3.1.5 PRIORITY

Low

### 3.2 ADD TO HOME SCREEN FEATURE

#### 3.2.1 DESCRIPTION

The system shall prompt users with an 'Add to Home Screen' option for installing the PWA on their mobile, and display a download icon in the URL bar to install the PWA on their desktop.

#### 3.2.2 SOURCE

Sponsor Meeting

#### 3.2.3 CONSTRAINTS

The app must include a valid manifest file and meet browser criteria for the prompt to appear.

#### 3.2.4 STANDARDS

Web App Manifest standards, PWA installation best practices

#### 3.2.5 PRIORITY

High

### 3.3 PUSH NOTIFICATIONS FOR REMINDERS

#### 3.3.1 DESCRIPTION

The system shall support push notifications, allowing users to receive reminders and updates about tasks when the PWA is not open.

### **3.3.2 SOURCE**

Customer Meeting

### **3.3.3 CONSTRAINTS**

Push notifications must be user-approved and comply with privacy laws such as GDPR.

### **3.3.4 STANDARDS**

W3C Push API, Web Notifications API

### **3.3.5 PRIORITY**

Critical

## **3.4 MOBILE-FIRST RESPONSIVE DESIGN**

### **3.4.1 DESCRIPTION**

The system shall prioritize mobile-first design, ensuring optimal performance and usability on small screens.

### **3.4.2 SOURCE**

Sponsor Meeting

### **3.4.3 CONSTRAINTS**

The design must adapt to various screen sizes, ensuring usability on both small and large screens.

### **3.4.4 STANDARDS**

Responsive Web Design (RWD) best practices

### **3.4.5 PRIORITY**

Critical

## **3.5 DATA SYNC ACROSS DEVICES**

### **3.5.1 DESCRIPTION**

The system shall ensure that user data is synced across devices when users log into their accounts.

### **3.5.2 SOURCE**

Customer Meeting

### **3.5.3 CONSTRAINTS**

Data should be stored locally using IndexedDB for offline access and synced when a connection is available.

### **3.5.4 STANDARDS**

Firebase Firestore synchronization, IndexedDB

### **3.5.5 PRIORITY**

High

## **3.6 MANUAL REFRESHING**

### **3.6.1 DESCRIPTION**

When the PWA is installed on a mobile device, users shall be able to manually refresh the page by pulling down on the screen, as is standard in most mobile apps and websites. This interaction must trigger a page reload or content update.

### **3.6.2 SOURCE**

Customer Meeting

### **3.6.3 CONSTRAINTS**

The pull-to-refresh action must not interfere with other touch gestures on the page. It should provide visual feedback (e.g., loading spinner) to indicate that a refresh is in progress, ensuring a seamless user experience.

### **3.6.4 STANDARDS**

Mobile Web Interaction Standards, Pull-to-Refresh UI Pattern

### **3.6.5 PRIORITY**

Moderate

## **3.7 SET NOTIFICATION PREFERENCES**

### **3.7.1 DESCRIPTION**

Users shall be able to toggle the notification system on or off and choose between Weekly, Daily, and Urgent notifications. Changes are saved via the 'Update Preferences' button.

### **3.7.2 SOURCE**

Customer Meeting

### **3.7.3 CONSTRAINTS**

Preferences must persist across sessions. The user interface must be accessible to users with disabilities.

### **3.7.4 STANDARDS**

UI/UX best practices, WCAG 2.1 guidelines (Form controls, User preferences)

### **3.7.5 PRIORITY**

Critical

## **3.8 UPDATED PREFERENCE MESSAGE**

### **3.8.1 DESCRIPTION**

After clicking 'Update Preferences,' users will see a message indicating the success or failure of saving their preferences.

### **3.8.2 SOURCE**

Customer Meeting

### **3.8.3 CONSTRAINTS**

Success and error states must be visually distinct and accessible.

### **3.8.4 STANDARDS**

UI/UX standards for user feedback

### **3.8.5 PRIORITY**

High

## **3.9 WEEKLY NOTIFICATIONS**

### **3.9.1 DESCRIPTION**

Every Monday, users with Weekly notifications enabled will receive an email listing tasks that are past due or due in the next 7 days, or a 'Nothing due' message.

### **3.9.2 SOURCE**

Customer Meeting

### **3.9.3 CONSTRAINTS**

Emails must comply with CAN-SPAM and be sent from a verified domain using SPF and DKIM for security.

### **3.9.4 STANDARDS**

SendGrid API guidelines, WCAG 2.1 (Accessible email content), SPF, DKIM, CAN-SPAM Act

### **3.9.5 PRIORITY**

High

## **3.10 DAILY NOTIFICATIONS**

### **3.10.1 DESCRIPTION**

Daily notifications provide an email listing tasks which are overdue or due in the next 2 days, or a 'Nothing due' message if there are no tasks.

### **3.10.2 SOURCE**

Customer Meeting

### **3.10.3 CONSTRAINTS**

Ensure emails include an accessible 'Unsubscribe' link and comply with security protocols for email integrity.

### **3.10.4 STANDARDS**

SendGrid API guidelines, Accessible email formatting, SPF, DKIM, CAN-SPAM Act

### **3.10.5 PRIORITY**

High

## **3.11 URGENT NOTIFICATIONS**

### **3.11.1 DESCRIPTION**

Urgent notifications are sent for tasks that are overdue or due within the next 24 hours.

### **3.11.2 SOURCE**

Customer Meeting

### **3.11.3 CONSTRAINTS**

Urgent notifications must be sent in near-real-time with minimal delay.

### **3.11.4 STANDARDS**

Real-time notification standards, SendGrid API, Firebase Cloud Functions

### **3.11.5 PRIORITY**

Critical

## **3.12 NOTIFICATION CONTENT**

### **3.12.1 DESCRIPTION**

Task notification emails include Subject, Assignment, Due Date, and Project, formatted with HTML. Missing fields are omitted. Tasks are separated with \br tag.

### **3.12.2 SOURCE**

Customer Meeting

### **3.12.3 CONSTRAINTS**

The email template must be responsive and accessible to users with visual impairments.

### **3.12.4 STANDARDS**

HTML5 standards for email formatting, WCAG guidelines for email content

### **3.12.5 PRIORITY**

Moderate

## **3.13 NOTIFICATION FORMATTING**

### **3.13.1 DESCRIPTION**

The email templates must use the LearnLeaf Organizer branding, including Dark Green (#355147), Muted Brown (#9F6C5B), and Teal (#5B8E9F). The LearnLeaf logo should be displayed prominently at the top of the email. Additionally, the template must include a button labeled 'Go to Tasks' that directs users to the PWA URL for task management.

### **3.13.2 SOURCE**

Customer Meeting

### **3.13.3 CONSTRAINTS**

Bulk emails must comply with Gmail and SendGrid standards, ensuring the color palette is accessible and meets contrast ratio guidelines. The PWA button must be accessible and visible.

### **3.13.4 STANDARDS**

SendGrid template styling, WCAG 2.1 for email accessibility, Gmail bulk email standards

### **3.13.5 PRIORITY**

High

### **3.14 PROJECT PAGE THEME**

#### **3.14.1 DESCRIPTION**

The GUI shall use the following colors: light gray (#F3F2F2) for the overall background. The project-related components use dark green (#355147) for the project name and muted brown (#9F6C5B) for due dates and next tasks.

#### **3.14.2 SOURCE**

Customer Meeting

#### **3.14.3 CONSTRAINTS**

All specified colors (e.g., #F3F2F2 for background, #355147 for text, #9F6C5B for task details) must meet WCAG standards for accessibility, particularly in terms of contrast ratios.

#### **3.14.4 STANDARDS**

CSS3 and customer-provided color guidelines

#### **3.14.5 PRIORITY**

Moderate

### **3.15 PROJECT WIDGET THEME**

#### **3.15.1 DESCRIPTION**

The project widgets shall have a light gray background (#F3F2F2), with a border color of light gray (#DDDDDD). The project name text shall use dark green (#355147), while the due date and next task details shall use muted brown (#9F6C5B). Fonts shall be customer-approved and consistent across the application, ensuring readability.

#### **3.15.2 SOURCE**

Customer Meeting

#### **3.15.3 CONSTRAINTS**

The font sizes and color schemes must comply with accessibility standards to ensure readability.

#### **3.15.4 STANDARDS**

CSS3 font and color standards, WCAG guidelines for accessibility

#### **3.15.5 PRIORITY**

Moderate

### **3.16 PROJECT WIDGET DOUGHNUT CHART**

#### **3.16.1 DESCRIPTION**

The doughnut chart shall represent project completion statuses with color-coding: #4CAF50 for completed tasks, #FF9800 for in-progress tasks, #FFC107 for pending tasks, and #F44336 for overdue tasks. The chart shall be hoverable, providing detailed information when hovered over.

#### **3.16.2 SOURCE**

Customer Meeting

### **3.16.3 CONSTRAINTS**

The color-coding (e.g., #4CAF50 for completed tasks, #FF9800 for in-progress, #FFC107 for pending, and #F44336 for overdue) must be distinguishable for users with color blindness. Hoverable elements must work across devices.

### **3.16.4 STANDARDS**

AnyChart library for data visualization, CSS3 for hover effects

### **3.16.5 PRIORITY**

High

## **3.17 PROJECT TASKS**

### **3.17.1 DESCRIPTION**

When a user clicks on a project name, the system shall redirect them to the task list, with tasks filtered to only show tasks assigned to that project.

### **3.17.2 SOURCE**

Customer Meeting

### **3.17.3 CONSTRAINTS**

The filtering mechanism must function efficiently even with large task lists, and navigation should be seamless.

### **3.17.4 STANDARDS**

React.js and routing best practices

### **3.17.5 PRIORITY**

Critical

## **3.18 PROJECT DELETION**

### **3.18.1 DESCRIPTION**

When a project is deleted, the associated tasks should remain in the system, linked to other relevant contexts but removed from the project view.

### **3.18.2 SOURCE**

Customer Meeting

### **3.18.3 CONSTRAINTS**

Associated tasks must remain accessible and be linked to other projects or contexts if necessary.

### **3.18.4 STANDARDS**

Data persistence best practices

### **3.18.5 PRIORITY**

Critical

### **3.19 PROJECT DELETE BUTTON THEME**

#### **3.19.1 DESCRIPTION**

Each project widget shall include a delete button styled with a red background (#F44336) and white text (#FFFFFF). Upon clicking, a confirmation prompt shall appear to prevent accidental deletion.

#### **3.19.2 SOURCE**

Customer Meeting

#### **3.19.3 CONSTRAINTS**

Project deletion should not impact associated tasks or data integrity, requiring careful state management.

#### **3.19.4 STANDARDS**

UI/UX confirmation standards

#### **3.19.5 PRIORITY**

Critical

### **3.20 PROJECT WIDGET TOOLTIP**

#### **3.20.1 DESCRIPTION**

Shows a tooltip when hovering over the project name, with the message 'View tasks for this project'.

#### **3.20.2 SOURCE**

Customer Meeting

#### **3.20.3 CONSTRAINTS**

May not work in very old browsers without pseudo-element support.

#### **3.20.4 STANDARDS**

CSS3 Pseudo-elements

#### **3.20.5 PRIORITY**

Moderate

### **3.21 PROJECT WIDGET BUTTONS**

#### **3.21.1 DESCRIPTION**

Buttons in the project widget are arranged in a row using Flexbox, including an 'Archive' button.

#### **3.21.2 SOURCE**

Customer Meeting

#### **3.21.3 CONSTRAINTS**

Button size may affect usability on smaller screens.

#### **3.21.4 STANDARDS**

CSS3 Flexbox

#### **3.21.5 PRIORITY**

Moderate

## **3.22 FILTER PROJECTS BY DUE DATE**

### **3.22.1 DESCRIPTION**

Users should be able to filter projects based on the due date of each project.

### **3.22.2 SOURCE**

Customer Meeting

### **3.22.3 CONSTRAINTS**

Filtering based on due dates should account for tasks with no due dates as well.

### **3.22.4 STANDARDS**

React.js filtering best practices

### **3.22.5 PRIORITY**

High

## **3.23 FILTER PROJECTS BY SUBJECT**

### **3.23.1 DESCRIPTION**

Users should be able to filter projects by the subject the project is associated with.

### **3.23.2 SOURCE**

Customer Meeting

### **3.23.3 CONSTRAINTS**

The filter should support multiple subjects for projects with interdisciplinary scope.

### **3.23.4 STANDARDS**

React.js filtering best practices

### **3.23.5 PRIORITY**

High

## **3.24 SEARCH FOR PROJECTS BY NAME**

### **3.24.1 DESCRIPTION**

Users should be able to input a project name in the search bar to find specific projects.

### **3.24.2 SOURCE**

Customer Meeting

### **3.24.3 CONSTRAINTS**

Search should work efficiently even with large datasets.

### **3.24.4 STANDARDS**

React.js search and filtering best practices

### **3.24.5 PRIORITY**

High

### **3.25 FAB THEME**

#### **3.25.1 DESCRIPTION**

The form for adding new projects shall be accessed via a floating action button (FAB) located at the bottom right of the page. The FAB is styled with a teal background (#5B8E9F) and white text. On hover, the background changes to a darker blue (#0056b3).

#### **3.25.2 SOURCE**

Customer Meeting

#### **3.25.3 CONSTRAINTS**

Requires form validation and internet connectivity.

#### **3.25.4 STANDARDS**

React.js, JSX

#### **3.25.5 PRIORITY**

Critical

### **3.26 ADDING TASKS TO PROJECT**

#### **3.26.1 DESCRIPTION**

The form for adding tasks shall auto-populate the project name and subject name when adding tasks to an existing project.

#### **3.26.2 SOURCE**

Customer Meeting

#### **3.26.3 CONSTRAINTS**

Must validate project and subject match before auto-filling.

#### **3.26.4 STANDARDS**

React.js controlled components

#### **3.26.5 PRIORITY**

High

### **3.27 PROJECT ARCHIVE BUTTON THEME**

#### **3.27.1 DESCRIPTION**

The archive button shall be styled with a muted brown background (#BA7E6A) and black text. Clicking on the archive button moves projects to the Archives page.

#### **3.27.2 SOURCE**

Customer Meeting

#### **3.27.3 CONSTRAINTS**

Archived projects must remain retrievable.

#### **3.27.4 STANDARDS**

UI/UX standards for archiving

### **3.27.5 PRIORITY**

Moderate

## **3.28 PROJECT WIDGET CONTENT**

### **3.28.1 DESCRIPTION**

The project widgets shall display the due date, next task, and subject info in a visually distinct way.

### **3.28.2 SOURCE**

Customer Meeting

### **3.28.3 CONSTRAINTS**

Text contrast may need adjustment for accessibility.

### **3.28.4 STANDARDS**

WCAG accessibility standards

### **3.28.5 PRIORITY**

High

## **3.29 PROJECT DELETION**

### **3.29.1 DESCRIPTION**

The user shall be able to delete projects, removing them from the dashboard. The system shall ask for user confirmation when a user attempts to delete a project.

### **3.29.2 SOURCE**

Customer Meeting

### **3.29.3 CONSTRAINTS**

Deletion is permanent, confirmation prompt is required.

### **3.29.4 STANDARDS**

UI/UX standards

### **3.29.5 PRIORITY**

Critical

## **3.30 PROJECT TASK MANAGEMENT**

### **3.30.1 DESCRIPTION**

Users can add, edit, and delete tasks associated with a project.

### **3.30.2 SOURCE**

Customer Meeting

### **3.30.3 CONSTRAINTS**

Tasks must be properly linked and trigger updates across the dashboard.

### **3.30.4 STANDARDS**

React.js state management

### **3.30.5 PRIORITY**

Critical

## **3.31 PROJECT EDIT BUTTON THEME**

### **3.31.1 DESCRIPTION**

Each project widget shall include an edit button styled with a muted teal background (#5B8E9F) and white text (#FFFFFF). Clicking the edit button opens a form pre-populated with the project's current details, allowing users to modify the project name, subject, and due date.

### **3.31.2 SOURCE**

Customer Meeting

### **3.31.3 CONSTRAINTS**

Requires form validation and internet connectivity.

### **3.31.4 STANDARDS**

React.js, JSX

### **3.31.5 PRIORITY**

High

## **3.32 PROJECT MANUAL REORDERING**

### **3.32.1 DESCRIPTION**

Users shall be able to manually reorder projects on the dashboard by dragging and dropping widgets.

### **3.32.2 SOURCE**

Customer Meeting

### **3.32.3 CONSTRAINTS**

Drag-and-drop needs to work on various screen sizes.

### **3.32.4 STANDARDS**

JavaScript Drag-and-Drop API

### **3.32.5 PRIORITY**

Low

## **3.33 PROJECT DASHBOARD DYNAMIC DISPLAY**

### **3.33.1 DESCRIPTION**

The system shall modify the number of project widgets displayed per row according to the screen size: 3 widgets per row on large screens, 2 widgets per row on medium screens, and 1 widget per row on small screens. Additionally, the chart legend shall be eliminated on small screens to optimize space. This ensures that the layout remains user-friendly and accessible across various devices, including desktops, tablets, and mobile phones.

### **3.33.2 SOURCE**

Customer Meeting

### **3.33.3 CONSTRAINTS**

The layout must remain responsive and eliminate unnecessary elements, such as the chart legend, to maximize screen space on smaller devices. Accessibility and usability must be maintained across all screen sizes.

### **3.33.4 STANDARDS**

CSS media queries, Responsive web design principles, AnyChart customization for mobile

### **3.33.5 PRIORITY**

Moderate

## **3.34 REMEMBER-ME FUNCTIONALITY**

### **3.34.1 SOURCE**

Benjamin Niccum

### **3.34.2 DESCRIPTION**

The system shall offer a "Remember me" option to keep users logged in.

### **3.34.3 REQUIREMENTS AREA**

Customer

### **3.34.4 IMPORTANCE**

Moderate

### **3.34.5 APPLICABLE STANDARD**

OAuth 2.0, JWT

### **3.34.6 CONSTRAINTS**

Implement secure token storage and refresh mechanisms.

## **3.35 ERROR HANDLING AND FEEDBACK**

### **3.35.1 SOURCE**

Benjamin Niccum

### **3.35.2 DESCRIPTION**

The system shall provide clear, user-friendly error messages for login failures.

### **3.35.3 REQUIREMENTS AREA**

Customer

### **3.35.4 IMPORTANCE**

High

### **3.35.5 APPLICABLE STANDARD**

WCAG 2.1 Success Criterion 3.3.1

### **3.35.6 CONSTRAINTS**

Avoid revealing specific account information in error messages.

### **3.36 RESPONSIVE DESIGN FOR LOGIN PAGE**

#### **3.36.1 SOURCE**

Benjamin Niccum

#### **3.36.2 DESCRIPTION**

The login page shall be responsive and usable on various devices and screen sizes.

#### **3.36.3 REQUIREMENTS AREA**

Customer

#### **3.36.4 IMPORTANCE**

High

#### **3.36.5 APPLICABLE STANDARD**

Responsive Web Design principles

#### **3.36.6 CONSTRAINTS**

Ensure all elements are accessible and functional on mobile devices.

### **3.37 USER-FRIENDLY REGISTRATION FORM**

#### **3.37.1 SOURCE**

Benjamin Niccum

#### **3.37.2 DESCRIPTION**

The system shall provide a clear, easy-to-use registration form with necessary fields.

#### **3.37.3 REQUIREMENTS AREA**

Customer

#### **3.37.4 IMPORTANCE**

Critical

#### **3.37.5 APPLICABLE STANDARD**

WCAG 2.1

#### **3.37.6 CONSTRAINTS**

Form must be accessible and navigable via keyboard.

### **3.38 TERMS OF SERVICE AGREEMENT**

#### **3.38.1 SOURCE**

Benjamin Niccum

#### **3.38.2 DESCRIPTION**

The system shall require users to agree to terms of service before registration.

#### **3.38.3 REQUIREMENTS AREA**

Customer

#### **3.38.4 IMPORTANCE**

High

#### **3.38.5 APPLICABLE STANDARD**

GDPR, CCPA compliance

#### **3.38.6 CONSTRAINTS**

Ensure terms are easily accessible and written in clear language.

### **3.39 DATA PRIVACY NOTICE**

#### **3.39.1 SOURCE**

Benjamin Niccum

#### **3.39.2 DESCRIPTION**

The system shall provide a clear data privacy notice during registration.

#### **3.39.3 REQUIREMENTS AREA**

Customer

#### **3.39.4 IMPORTANCE**

Critical

#### **3.39.5 APPLICABLE STANDARD**

GDPR, CCPA compliance

#### **3.39.6 CONSTRAINTS**

Clearly explain data collection, usage, and user rights.

### **3.40 SOCIAL MEDIA SIGN-UP OPTION**

#### **3.40.1 SOURCE**

Customer Meeting

#### **3.40.2 DESCRIPTION**

The system shall allow registration via popular social media platforms.

#### **3.40.3 REQUIREMENTS AREA**

Customer

#### **3.40.4 IMPORTANCE**

Moderate

#### **3.40.5 APPLICABLE STANDARD**

OAuth 2.0, OpenID Connect

#### **3.40.6 CONSTRAINTS**

Comply with each platform's data usage and privacy policies.

### **3.41 PROGRESSIVE FORM FILLING**

#### **3.41.1 SOURCE**

Customer Meeting

#### **3.41.2 DESCRIPTION**

The system shall implement a multi-step registration process for complex sign-ups.

### **3.41.3 REQUIREMENTS AREA**

Customer

### **3.41.4 IMPORTANCE**

Moderate

### **3.41.5 APPLICABLE STANDARD**

UX best practices

### **3.41.6 CONSTRAINTS**

Ensure each step is saved to prevent data loss if user navigates away.

CUSTOMER

## **3.42 PUSH NOTIFICATION SYSTEM**

### **3.42.1 SOURCE**

Benjamin Niccum

### **3.42.2 DESCRIPTION**

The mobile app shall implement a push notification system to alert users about tasks, deadlines, and updates.

### **3.42.3 REQUIREMENTS AREA**

Customer

### **3.42.4 IMPORTANCE**

Critical

### **3.42.5 APPLICABLE STANDARD**

FCM (Android), APNs (iOS)

### **3.42.6 CONSTRAINTS**

Must handle scenarios where push notifications are disabled by the user.

## **3.43 BIOMETRIC AUTHENTICATION**

### **3.43.1 SOURCE**

Customer Meeting

### **3.43.2 DESCRIPTION**

The app shall support biometric authentication (fingerprint, face recognition) for quick and secure login.

### **3.43.3 REQUIREMENTS AREA**

Security

### **3.43.4 IMPORTANCE**

High

### **3.43.5 APPLICABLE STANDARD**

Android Biometric API, iOS Local Authentication framework

#### **3.43.6 CONSTRAINTS**

Fall back to password authentication when biometrics are unavailable.

### **3.44 CAMERA INTEGRATION**

#### **3.44.1 SOURCE**

Customer Meeting

#### **3.44.2 DESCRIPTION**

The app shall allow users to attach photos to tasks using the device's camera.

#### **3.44.3 REQUIREMENTS AREA**

Customer

#### **3.44.4 IMPORTANCE**

Moderate

#### **3.44.5 APPLICABLE STANDARD**

Android Camera2 API, iOS AVFoundation

#### **3.44.6 CONSTRAINTS**

Handle permissions and storage efficiently.

### **3.45 DEEP LINKING**

#### **3.45.1 SOURCE**

Customer Meeting

#### **3.45.2 DESCRIPTION**

The app shall support deep linking, allowing users to open specific tasks or views directly from notifications or external links.

#### **3.45.3 REQUIREMENTS AREA**

Customer

#### **3.45.4 IMPORTANCE**

High

#### **3.45.5 APPLICABLE STANDARD**

Android App Links, iOS Universal

#### **3.45.6 CONSTRAINTS**

Ensure proper handling of invalid or expired links.

### **3.46 HAPTIC FEEDBACK**

#### **3.46.1 SOURCE**

Benjamin Niccum

#### **3.46.2 DESCRIPTION**

The app shall provide haptic feedback for key actions such as completing tasks or setting reminders.

### **3.46.3 REQUIREMENTS AREA**

Customer

### **3.46.4 IMPORTANCE**

Moderate

### **3.46.5 APPLICABLE STANDARD**

Android Vibrator API, iOS Core Haptics

### **3.46.6 CONSTRAINTS**

Ensure subtle and non-intrusive feedback.

## **3.47 PUSH NOTIFICATION CHANNELS**

### **3.47.1 SOURCE**

Customer Meeting

### **3.47.2 DESCRIPTION**

The app shall implement notification channels/categories to allow users fine-grained control over notification types.

### **3.47.3 REQUIREMENTS AREA**

Customer

### **3.47.4 IMPORTANCE**

High

### **3.47.5 APPLICABLE STANDARD**

Android Notification Channels, iOS Notification Categories

### **3.47.6 CONSTRAINTS**

Provide clear descriptions for each notification channel/category.

## **3.48 MOBILE-SPECIFIC UI/UX**

### **3.48.1 SOURCE**

Benjamin Niccum

### **3.48.2 DESCRIPTION**

The app shall have a mobile-optimized interface with touch-friendly elements.

### **3.48.3 REQUIREMENTS AREA**

Customer

### **3.48.4 IMPORTANCE**

Critical

### **3.48.5 APPLICABLE STANDARD**

Material Design (Android), Human Interface Guidelines (iOS)

### **3.48.6 CONSTRAINTS**

Ensure consistency with platform-specific design patterns.

### **3.49 REACTIVATE ARCHIVED ITEMS**

#### **3.49.1 SOURCE**

User Experience Designer

#### **3.49.2 DESCRIPTION**

The system shall allow users to reactivate archived tasks, subjects, and projects.

#### **3.49.3 REQUIREMENTS AREA**

Customer

#### **3.49.4 IMPORTANCE**

Medium

#### **3.49.5 APPLICABLE STANDARD**

WCAG 2.1

#### **3.49.6 CONSTRAINTS**

Ensure clear user interface for reactivation process.

### **3.50 EDIT ARCHIVED ITEMS**

#### **3.50.1 SOURCE**

User Experience Designer

#### **3.50.2 DESCRIPTION**

The system shall allow users to edit tasks, subjects, and projects from the archive page.

#### **3.50.3 REQUIREMENTS AREA**

Customer

#### **3.50.4 IMPORTANCE**

Medium

#### **3.50.5 APPLICABLE STANDARD**

WCAG 2.1

#### **3.50.6 CONSTRAINTS**

Maintain data integrity during editing process.

### **3.51 DELETE ARCHIVED ITEMS**

#### **3.51.1 SOURCE**

User Experience Designer

#### **3.51.2 DESCRIPTION**

The system shall allow users to delete tasks, subjects, and projects from the archive page.

#### **3.51.3 REQUIREMENTS AREA**

Customer

#### **3.51.4 IMPORTANCE**

Medium

### **3.51.5 APPLICABLE STANDARD**

WCAG 2.1

### **3.51.6 CONSTRAINTS**

Implement confirmation step before permanent deletion.

## **3.52 TOGGLE ARCHIVE VISIBILITY**

### **3.52.1 SOURCE**

User Experience Designer

### **3.52.2 DESCRIPTION**

The system shall allow users to toggle the visibility of archived tasks, subjects, and projects.

### **3.52.3 REQUIREMENTS AREA**

Customer

### **3.52.4 IMPORTANCE**

Low

### **3.52.5 APPLICABLE STANDARD**

WCAG 2.1

### **3.52.6 CONSTRAINTS**

Ensure smooth transition between visibility states.

## **3.53 FILTER TASKS**

### **3.53.1 SOURCE**

User Experience Designer

### **3.53.2 DESCRIPTION**

The system shall allow users to filter tasks by assignment name, priority, status, due date, and start date.

### **3.53.3 REQUIREMENTS AREA**

Customer

### **3.53.4 IMPORTANCE**

High

### **3.53.5 APPLICABLE STANDARD**

WCAG 2.1

### **3.53.6 CONSTRAINTS**

Optimize filter performance for large datasets.

## **3.54 TASK PAGE TITLE**

### **3.54.1 SOURCE**

Learnleaf Organizer Website

#### **3.54.2 DESCRIPTION**

The Task page shall have a title which includes the user's name. The title shall display as User Name's Upcoming Tasks with placeholders replaced by name the user set in their profile.

#### **3.54.3 REQUIREMENTS AREA**

Customer

#### **3.54.4 IMPORTANCE**

Moderate

#### **3.54.5 APPLICABLE STANDARD**

UI/UX best practices for personalization

#### **3.54.6 CONSTRAINTS**

Title format must dynamically update based on the logged-in user's data.

### **3.55 NEW TASK MENU BUTTON**

#### **3.55.1 SOURCE**

Learnleaf Organizer Website

#### **3.55.2 DESCRIPTION**

The task page shall have a button which opens the New Task Menu. Users shall be able to click a button labeled "+" to open the New Task Menu.

#### **3.55.3 REQUIREMENTS AREA**

Customer

#### **3.55.4 IMPORTANCE**

High

#### **3.55.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

#### **3.55.6 CONSTRAINTS**

The button must be clearly labeled and accessible on all devices.

### **3.56 TASK TITLE INPUT**

#### **3.56.1 SOURCE**

Learnleaf Organizer Website

#### **3.56.2 DESCRIPTION**

The New Task Menu shall have a text field for entering the task's title. The New Task Menu shall include a required text field for the task title, preventing task creation without this field being filled.

#### **3.56.3 REQUIREMENTS AREA**

Customer

#### **3.56.4 IMPORTANCE**

Critical

### **3.56.5 APPLICABLE STANDARD**

Form validation standards

### **3.56.6 CONSTRAINTS**

Title input must not be left blank and should handle special characters.

## **3.57 TASK DESCRIPTION INPUT**

### **3.57.1 SOURCE**

Learnleaf Organizer Website

### **3.57.2 DESCRIPTION**

The New Task Menu shall have a multi-line text field for entering the task's description. The New Task Menu shall restrict the viewing of the description to 4 rows, with a scroll bar allowing users to view text below.

### **3.57.3 REQUIREMENTS AREA**

Customer

### **3.57.4 IMPORTANCE**

Low

### **3.57.5 APPLICABLE STANDARD**

HTML5 text area standards, UI/UX standards for multi-line text input fields

### **3.57.6 CONSTRAINTS**

The description text field must be limited to a maximum of 4 rows in height to maintain a compact layout, especially on mobile devices. A vertical scroll bar shall be included to facilitate viewing of extended text without expanding the field size. The field should allow a reasonable maximum character count to accommodate detailed descriptions without causing layout issues.

## **3.58 TASK SUBJECT SELECTION**

### **3.58.1 SOURCE**

Learnleaf Organizer Website

### **3.58.2 DESCRIPTION**

The New Task Menu shall have a dropdown for selecting the task's subject. Users can select a subject from a dropdown list that includes "None," "Add New", and any existing subjects.

### **3.58.3 REQUIREMENTS AREA**

Customer

### **3.58.4 IMPORTANCE**

High

### **3.58.5 APPLICABLE STANDARD**

UI/UX dropdown list standards

### **3.58.6 CONSTRAINTS**

Dropdown must dynamically update with current subjects and allow for new additions.

### **3.59 TASK PROJECT SELECTION**

#### **3.59.1 SOURCE**

Learnleaf Organizer Website

#### **3.59.2 DESCRIPTION**

The New Task Menu shall have a dropdown for selecting the task's project. Users can select a project from a dropdown list that includes "None," "Add New", and any existing projects.

#### **3.59.3 REQUIREMENTS AREA**

Customer

#### **3.59.4 IMPORTANCE**

High

#### **3.59.5 APPLICABLE STANDARD**

UI/UX dropdown list standards

#### **3.59.6 CONSTRAINTS**

The project list must update dynamically and support new projects.

### **3.60 TASK PRIORITY SELECTION**

#### **3.60.1 SOURCE**

Learnleaf Organizer Website

#### **3.60.2 DESCRIPTION**

The New Task Menu shall have a dropdown for selecting the task's priority. The priority dropdown shall include options "Low," "Medium," and "High."

#### **3.60.3 REQUIREMENTS AREA**

Customer

#### **3.60.4 IMPORTANCE**

Moderate

#### **3.60.5 APPLICABLE STANDARD**

UI/UX dropdown list standards

#### **3.60.6 CONSTRAINTS**

The dropdown options must be clearly labeled and accessible.

### **3.61 TASK STATUS SELECTION**

#### **3.61.1 SOURCE**

Learnleaf Organizer Website

#### **3.61.2 DESCRIPTION**

The New Task Menu shall have a dropdown for selecting the task's status. The status dropdown shall include options "Not Started," "In Progress," and "Completed."

#### **3.61.3 REQUIREMENTS AREA**

Customer

#### **3.61.4 IMPORTANCE**

Moderate

#### **3.61.5 APPLICABLE STANDARD**

UI/UX dropdown list standards

#### **3.61.6 CONSTRAINTS**

The dropdown options should be displayed consistently across all devices.

### **3.62 TASK START DATE SELECTION**

#### **3.62.1 SOURCE**

Learnleaf Organizer Website

#### **3.62.2 DESCRIPTION**

The New Task Menu shall have a date field for selecting the start date. Users shall be able to choose a start date using a date picker in the New Task Menu.

#### **3.62.3 REQUIREMENTS AREA**

Customer

#### **3.62.4 IMPORTANCE**

High

#### **3.62.5 APPLICABLE STANDARD**

Date picker UI standards

#### **3.62.6 CONSTRAINTS**

The date picker must support date formats compatible with local settings.

### **3.63 TASK DUE DATE SELECTION**

#### **3.63.1 SOURCE**

Learnleaf Organizer Website

#### **3.63.2 DESCRIPTION**

The New Task Menu shall have a date field for selecting the due date. Users shall be able to select a due date using a date picker.

#### **3.63.3 REQUIREMENTS AREA**

Customer

#### **3.63.4 IMPORTANCE**

High

#### **3.63.5 APPLICABLE STANDARD**

Date picker UI standards

#### **3.63.6 CONSTRAINTS**

The date picker must support date formats compatible with local settings.

### **3.64 TASK DUE TIME SELECTION**

#### **3.64.1 SOURCE**

Learnleaf Organizer Website

#### **3.64.2 DESCRIPTION**

The New Task Menu shall have a time field for selecting the due time. The time picker shall adjust to the user's locale, supporting 12-hour or 24-hour formats as appropriate.

#### **3.64.3 REQUIREMENTS AREA**

Customer

#### **3.64.4 IMPORTANCE**

High

#### **3.64.5 APPLICABLE STANDARD**

Time picker standards, Locale handling

#### **3.64.6 CONSTRAINTS**

The time picker must adjust based on device settings and user preferences.

### **3.65 NEW TASK MENU BUTTONS**

#### **3.65.1 SOURCE**

Learnleaf Organizer Website

#### **3.65.2 DESCRIPTION**

The New Task Menu shall have a button for canceling and a button for saving. The Cancel button shall close the New Task Menu without saving, while the Save button shall create a new task with the entered details.

#### **3.65.3 REQUIREMENTS AREA**

Customer

#### **3.65.4 IMPORTANCE**

Critical

#### **3.65.5 APPLICABLE STANDARD**

UI/UX standards for form submission and cancellation

#### **3.65.6 CONSTRAINTS**

The buttons must be distinguishable, with Save confirming task creation and Cancel closing the menu without changes.

### **3.66 TASK VIEWING AND EDITING**

#### **3.66.1 SOURCE**

Learnleaf Organizer Website

#### **3.66.2 DESCRIPTION**

The user shall be able to view and edit tasks on the Task page. Users must be able to directly interact with tasks, modifying details without opening a separate menu.

### **3.66.3 REQUIREMENTS AREA**

Customer

### **3.66.4 IMPORTANCE**

High

### **3.66.5 APPLICABLE STANDARD**

In-place editing UI standards

### **3.66.6 CONSTRAINTS**

The in-place editing feature must allow safe modifications with real-time updates.

## **3.67 TASK DISPLAY FORMAT**

### **3.67.1 SOURCE**

Learnleaf Organizer Website

### **3.67.2 DESCRIPTION**

Created tasks shall be listed as a block in a grid style. The task list shall be displayed in a responsive grid format that adapts to the screen size.

### **3.67.3 REQUIREMENTS AREA**

Customer

### **3.67.4 IMPORTANCE**

Moderate

### **3.67.5 APPLICABLE STANDARD**

Responsive web design standards

### **3.67.6 CONSTRAINTS**

The grid must dynamically adjust based on screen size and orientation for optimal viewing.

## **3.68 TASK BLOCK COMPONENTS**

### **3.68.1 SOURCE**

Learnleaf Organizer Website

### **3.68.2 DESCRIPTION**

Each task block shall display all of the same components and have the same functionality as the New Task Menu except for Description. Each task block shall display editable task details, except for the Title (non-editable) and a Delete button in place of Cancel. Deleting a task removes it from the list.

### **3.68.3 REQUIREMENTS AREA**

Customer

### **3.68.4 IMPORTANCE**

Critical

### **3.68.5 APPLICABLE STANDARD**

UI/UX editing and deletion standards

### **3.68.6 CONSTRAINTS**

The Delete button must prompt for confirmation and remove the task from both the list and database upon confirmation.

## **3.69 TASK EDITING INITIATION**

### **3.69.1 SOURCE**

Learnleaf Organizer Website

### **3.69.2 DESCRIPTION**

The user shall be able to edit a task by clicking the edit icon on the desired task. Users can initiate task editing by clicking the edit icon on the task they wish to modify. This icon should be easily recognizable and consistently placed next to each task.

### **3.69.3 REQUIREMENTS AREA**

Customer

### **3.69.4 IMPORTANCE**

High

### **3.69.5 APPLICABLE STANDARD**

UI/UX standards for edit actions

### **3.69.6 CONSTRAINTS**

The edit icon must be visible and accessible on all interfaces.

## **3.70 EDIT TASK FORM DISPLAY**

### **3.70.1 SOURCE**

Learnleaf Organizer Website

### **3.70.2 DESCRIPTION**

The system shall display an Edit Task Form after the edit icon is clicked. After clicking the edit icon, the system shall display the Edit Task Form as an overlay.

### **3.70.3 REQUIREMENTS AREA**

Customer

### **3.70.4 IMPORTANCE**

High

### **3.70.5 APPLICABLE STANDARD**

UI/UX standards for modal forms

### **3.70.6 CONSTRAINTS**

The form should load quickly and be usable across all devices.

## **3.71 EDIT TASK FORM PRE-FILL**

### **3.71.1 SOURCE**

Learnleaf Organizer Website

### **3.71.2 DESCRIPTION**

The system shall automatically load the task details into the Edit Task Form. The system shall pre-fill the form with the task's existing details to allow for easy modification.

### **3.71.3 REQUIREMENTS AREA**

Customer

### **3.71.4 IMPORTANCE**

High

### **3.71.5 APPLICABLE STANDARD**

React.js state management, Pre-filled form standards

### **3.71.6 CONSTRAINTS**

The details must be accurate and consistent with the database.

## **3.72 EDIT TASK FORM FUNCTIONALITY**

### **3.72.1 SOURCE**

Learnleaf Organizer Website

### **3.72.2 DESCRIPTION**

The user shall be able to edit and set the task details in the same way as the Add Task Form. Users can modify fields in the Edit Task Form, with input and validation mirroring the Add Task Form.

### **3.72.3 REQUIREMENTS AREA**

Customer

### **3.72.4 IMPORTANCE**

Critical

### **3.72.5 APPLICABLE STANDARD**

React.js form handling, Validation standards

### **3.72.6 CONSTRAINTS**

The form must support identical fields and validation to the Add Task Form.

## **3.73 EDIT TASK FORM SAVE BUTTON**

### **3.73.1 SOURCE**

Learnleaf Organizer Website

### **3.73.2 DESCRIPTION**

The system shall display a Save button on the Edit Task Form, which saves the updated task details, closes the form, and refreshes the list of all tasks. The Save button shall commit changes to the database, close the form, and refresh the task list.

### **3.73.3 REQUIREMENTS AREA**

Customer

### **3.73.4 IMPORTANCE**

Critical

### **3.73.5 APPLICABLE STANDARD**

React.js state management, Form submission UI/UX standards

### **3.73.6 CONSTRAINTS**

Save action must confirm changes and update the interface without errors.

## **3.74 EDIT TASK FORM CANCEL BUTTON**

### **3.74.1 SOURCE**

Learnleaf Organizer Website

### **3.74.2 DESCRIPTION**

The system shall display a Cancel button, which simply closes the form without modifying the task. The Cancel button allows users to exit the form without saving changes.

### **3.74.3 REQUIREMENTS AREA**

Customer

### **3.74.4 IMPORTANCE**

High

### **3.74.5 APPLICABLE STANDARD**

UI/UX standards for cancel actions

### **3.74.6 CONSTRAINTS**

The Cancel button must not alter task data and should be distinguishable from the Save button.

## **3.75 SUBJECT PAGE TITLE**

### **3.75.1 SOURCE**

Learnleaf Organizer Website

### **3.75.2 DESCRIPTION**

The Subject page shall have a title which includes the user's name. It should display as "FirstName LastName's Current Subjects" where FirstName and LastName are the user's first and last name respectively. (display just username)

### **3.75.3 REQUIREMENTS AREA**

Customer

### **3.75.4 IMPORTANCE**

3

### **3.75.5 APPLICABLE STANDARD**

UI/UX best practices for personalization

### **3.75.6 CONSTRAINTS**

Title format must dynamically update based on the logged-in user's data.

## **3.76 VIEW AND EDIT SUBJECT**

### **3.76.1 SOURCE**

Learnleaf Organizer Website

### **3.76.2 DESCRIPTION**

The user shall be able to view and edit subjects on the Subject page.

### **3.76.3 REQUIREMENTS AREA**

Customer

### **3.76.4 IMPORTANCE**

1

### **3.76.5 APPLICABLE STANDARD**

UI/UX best practices for personalization

### **3.76.6 CONSTRAINTS**

The button must be clearly labeled and accessible on all devices.

## **3.77 CREATING SUBJECTS**

### **3.77.1 SOURCE**

Learnleaf Organizer Website

### **3.77.2 DESCRIPTION**

Created subjects shall be listed as a block in a grid style. The block should display the subject's color on its border.

### **3.77.3 REQUIREMENTS AREA**

Customer

### **3.77.4 IMPORTANCE**

3

### **3.77.5 APPLICABLE STANDARD**

UI/UX best practices for personalization

### **3.77.6 CONSTRAINTS**

Title input must not be left blank and should handle special characters.

## **3.78 SUBJECT LIST FORMAT**

### **3.78.1 SOURCE**

Learnleaf Organizer Website

### **3.78.2 DESCRIPTION**

Each subject block shall display the subject's title.

### **3.78.3 REQUIREMENTS AREA**

Customer

### **3.78.4 IMPORTANCE**

2

### **3.78.5 APPLICABLE STANDARD**

UI/UX best practices for personalization

### **3.78.6 CONSTRAINTS**

Dropdown must dynamically update with current subjects and allow for new additions.

## **3.79 SUBJECT EDIT MENU**

### **3.79.1 SOURCE**

Learnleaf Organizer Website

### **3.79.2 DESCRIPTION**

Each subject block shall have a button that opens the edit menu.

### **3.79.3 REQUIREMENTS AREA**

Customer

### **3.79.4 IMPORTANCE**

2

### **3.79.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

### **3.79.6 CONSTRAINTS**

The edit icon must be visible and accessible on all interfaces.

## **3.80 SUBJECT DELETION**

### **3.80.1 SOURCE**

Learnleaf Organizer Website

### **3.80.2 DESCRIPTION**

Each subject shall have a button for deleting. The user shall be prompted to confirm deletion.

### **3.80.3 REQUIREMENTS AREA**

Customer

### **3.80.4 IMPORTANCE**

2

### **3.80.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

### **3.80.6 CONSTRAINTS**

The dropdown options must be clearly labeled and accessible.

## **3.81 SUBJECT ARCHIVING**

### **3.81.1 SOURCE**

Learnleaf Organizer Website

### **3.81.2 DESCRIPTION**

Each subject shall have a button for archiving. Archiving a subject shall remove it from the Subject page and add it to the subject archive page.

### **3.81.3 REQUIREMENTS AREA**

Customer

### **3.81.4 IMPORTANCE**

3

### **3.81.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

### **3.81.6 CONSTRAINTS**

The dropdown options should be displayed consistently across all devices.

## **3.82 SUBJECT EDITING**

### **3.82.1 SOURCE**

Learnleaf Organizer Website

### **3.82.2 DESCRIPTION**

The edit subject menu shall allow the user to change the subject's title, semester, description, and color. The subject title is a required field for creating a new subject. The user shall have the option to change color by directly enter a hexadecimal value or by using a color picker. The subject's data shall be preloaded in to the fields.

### **3.82.3 REQUIREMENTS AREA**

Customer

### **3.82.4 IMPORTANCE**

3

### **3.82.5 APPLICABLE STANDARD**

UI/UX color picker standards

### **3.82.6 CONSTRAINTS**

The date picker must support date formats compatible with local settings.

## **3.83 SAVING NEW SUBJECTS**

### **3.83.1 SOURCE**

Learnleaf Organizer Website

### **3.83.2 DESCRIPTION**

The edit subject menu shall have a save button which adds the subject to the subject list.

### **3.83.3 REQUIREMENTS AREA**

Customer

### **3.83.4 IMPORTANCE**

3

### **3.83.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

#### **3.83.6 CONSTRAINTS**

The date must be validated to ensure it is not before the start date.

### **3.84 SUBJECT CREATION CANCELING**

#### **3.84.1 SOURCE**

Learnleaf Organizer Website

#### **3.84.2 DESCRIPTION**

The edit subject menu shall have a cancel button which return to the subject page.

#### **3.84.3 REQUIREMENTS AREA**

Customer

#### **3.84.4 IMPORTANCE**

4

#### **3.84.5 APPLICABLE STANDARD**

UI/UX standards for cancel actions

#### **3.84.6 CONSTRAINTS**

The time picker must adjust based on device settings and user preferences.

### **3.85 SUBJECT FILTERING**

#### **3.85.1 SOURCE**

Learnleaf Organizer Website

#### **3.85.2 DESCRIPTION**

The user shall be able to filter the Subject list. Filter options shall be contained in a drop down panel which can be toggled open and closed.

#### **3.85.3 REQUIREMENTS AREA**

Customer

#### **3.85.4 IMPORTANCE**

1

#### **3.85.5 APPLICABLE STANDARD**

UI/UX standards for filter interfaces

#### **3.85.6 CONSTRAINTS**

The buttons must be distinguishable, with Save confirming task creation and Cancel closing the menu without changes.

### **3.86 FILTER SUBJECTS BY SUBJECT, SEMESTER, AND DESCRIPTION**

#### **3.86.1 SOURCE**

Learnleaf Organizer Website

### **3.86.2 DESCRIPTION**

The filter panel shall contain a search by subject field and a search by semester field. Filtering by subject or field should happen dynamically, meaning the subjects list should be updated each time a character is entered or delete from the text field.

### **3.86.3 REQUIREMENTS AREA**

Customer

### **3.86.4 IMPORTANCE**

1

### **3.86.5 APPLICABLE STANDARD**

UI/UX standards for filter interfaces

### **3.86.6 CONSTRAINTS**

Task data types must align with Firebase's data requirements for seamless storage.

## **3.87 CLEAR SUBJECT FILTERS**

### **3.87.1 SOURCE**

Learnleaf Organizer Website

### **3.87.2 DESCRIPTION**

The filter panel shall have a button which clears all currently applied filters.

### **3.87.3 REQUIREMENTS AREA**

Customer

### **3.87.4 IMPORTANCE**

4

### **3.87.5 APPLICABLE STANDARD**

UI/UX standards for cancel actions

### **3.87.6 CONSTRAINTS**

The in-place editing feature must allow safe modifications with real-time updates.

## **3.88 SUBJECT CREATION**

### **3.88.1 SOURCE**

Learnleaf Organizer Website

### **3.88.2 DESCRIPTION**

The subject page shall have a button for opening the new subjects menu. The button should be located at the bottom left corner of the screen.

### **3.88.3 REQUIREMENTS AREA**

Customer

### **3.88.4 IMPORTANCE**

3

### **3.88.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

### **3.88.6 CONSTRAINTS**

The grid must dynamically adjust based on screen size and orientation for optimal viewing.

## **3.89 CREATION MENU UNIFORMITY**

### **3.89.1 SOURCE**

Learnleaf Organizer Website

### **3.89.2 DESCRIPTION**

The new subject menu should be exactly the same as the edit subject menu. The display text of the save button shall be "ADD SUBJECT". The default color shall be Black.

### **3.89.3 REQUIREMENTS AREA**

Customer

### **3.89.4 IMPORTANCE**

2

### **3.89.5 APPLICABLE STANDARD**

UI/UX standards for modal menus

### **3.89.6 CONSTRAINTS**

The Delete button must prompt for confirmation and remove the task from both the list and database upon confirmation.

## **3.90 INTERNAL SUBJECT LINKS**

### **3.90.1 SOURCE**

Learnleaf Organizer Website

### **3.90.2 DESCRIPTION**

Clicking on a subject title shall redirect the user to the tasks page. When directing, filter by the selected task's subject.

### **3.90.3 REQUIREMENTS AREA**

Customer

### **3.90.4 IMPORTANCE**

2

### **3.90.5 APPLICABLE STANDARD**

JavaScript navigation standards, React.js component reusability best practices

### **3.90.6 CONSTRAINTS**

The filtering functionality must be pre-applied upon navigation to the tasks page, ensuring a smooth transition and immediate display of relevant tasks. The filter must accurately apply based on the clicked subject and handle cases where no tasks match the selected subject.

### **3.91 EDIT TASK FORM DISPLAY**

#### **3.91.1 SOURCE**

Learnleaf Organizer Website

#### **3.91.2 DESCRIPTION**

After clicking the edit icon, the system shall display the Edit Task Form as an overlay.

#### **3.91.3 REQUIREMENTS AREA**

Customer

#### **3.91.4 IMPORTANCE**

High

#### **3.91.5 APPLICABLE STANDARD**

UI/UX standards for modal forms

#### **3.91.6 CONSTRAINTS**

The form should load quickly and be usable across all devices.

### **3.92 PRE-FILL EDIT TASK FORM**

#### **3.92.1 SOURCE**

Learnleaf Organizer Website

#### **3.92.2 DESCRIPTION**

The system shall pre-fill the form with the task's existing details to allow for easy modification.

#### **3.92.3 REQUIREMENTS AREA**

Customer

#### **3.92.4 IMPORTANCE**

High

#### **3.92.5 APPLICABLE STANDARD**

React.js state management, Pre-filled form standards

#### **3.92.6 CONSTRAINTS**

The details must be accurate and consistent with the database.

### **3.93 EDIT TASK FORM FUNCTIONALITY**

#### **3.93.1 SOURCE**

Learnleaf Organizer Website

#### **3.93.2 DESCRIPTION**

Users can modify fields in the Edit Task Form, with input and validation mirroring the Add Task Form.

#### **3.93.3 REQUIREMENTS AREA**

Customer

#### **3.93.4 IMPORTANCE**

Critical

### **3.93.5 APPLICABLE STANDARD**

React.js form handling, Validation standards

### **3.93.6 CONSTRAINTS**

The form must support identical fields and validation to the Add Task Form.

## **3.94 SAVE UPDATED TASK**

### **3.94.1 SOURCE**

Learnleaf Organizer Website

### **3.94.2 DESCRIPTION**

The Save button shall commit changes to the database, close the form, and refresh the task list.

### **3.94.3 REQUIREMENTS AREA**

Customer

### **3.94.4 IMPORTANCE**

Critical

### **3.94.5 APPLICABLE STANDARD**

React.js state management, Form submission UI/UX standards

### **3.94.6 CONSTRAINTS**

Save action must confirm changes and refresh the interface without errors.

## **3.95 CANCEL EDIT TASK**

### **3.95.1 SOURCE**

Learnleaf Organizer Website

### **3.95.2 DESCRIPTION**

The Cancel button allows users to exit the form without saving changes.

### **3.95.3 REQUIREMENTS AREA**

Customer

### **3.95.4 IMPORTANCE**

High

### **3.95.5 APPLICABLE STANDARD**

UI/UX standards for cancel actions

### **3.95.6 CONSTRAINTS**

The Cancel button must not alter task data and should be distinguishable from the Save button.

## **3.96 CALENDAR COLOR SCHEMA REQUIREMENTS**

### **3.96.1 PRIORITY**

Moderate

### **3.96.2 DESCRIPTION**

The GUI shall use the following colors: light gray (F3F2F2) for the overall background. The project-related components use dark green (355147) for the project name and muted brown (9F6C5B) for due dates and next tasks.

### **3.96.3 SOURCE**

Customer

### **3.96.4 APPLICABLE STANDARDS**

CSS3 and customer-provided color guidelines

### **3.96.5 CONSTRAINTS**

All specified colors (e.g., F3F2F2 for background, 355147 for text, 9F6C5B for task details) must meet WCAG standards for accessibility, particularly in terms of contrast ratios.

## **3.97 CALENDAR WITH TASKS**

### **3.97.1 PRIORITY**

Moderate

### **3.97.2 DESCRIPTION**

The calendar shall have a white background, with a border color of light gray (DDDDDD). The date text shall use black. Fonts shall be customer-approved and consistent across the application, ensuring readability.

### **3.97.3 SOURCE**

Customer Meeting

### **3.97.4 APPLICABLE STANDARDS**

CSS3 and customer-provided color guidelines

### **3.97.5 CONSTRAINTS**

All specified colors must meet WCAG standards for accessibility, particularly in terms of contrast ratios.

## **3.98 TASK DISPLAY ON CALENDAR**

### **3.98.1 PRIORITY**

Moderate

### **3.98.2 DESCRIPTION**

The name of each task will be highlighted in the color corresponding to the subject that it is assigned to

### **3.98.3 SOURCE**

Customer Meeting

### **3.98.4 APPLICABLE STANDARDS**

React.js best coding practices

### **3.98.5 CONSTRAINTS**

All specified colors must meet WCAG standards for accessibility, particularly in terms of contrast ratios.

### **3.99 CALENDAR DATE RANGE**

#### **3.99.1 PRIORITY**

Moderate

#### **3.99.2 DESCRIPTION**

The user will have button options that allow them to select a date range. The display will change without requiring reload of the page.

#### **3.99.3 SOURCE**

Customer Meeting

#### **3.99.4 APPLICABLE STANDARDS**

React.js best coding practices

#### **3.99.5 CONSTRAINTS**

The font sizes and color schemes must comply with accessibility standards to ensure readability.

### **3.100 CALENDAR VIEW OPTIONS**

#### **3.100.1 PRIORITY**

Moderate

#### **3.100.2 DESCRIPTION**

Options for a month, week and day must be provided. The month view will have days organized increasing in number from left to right and top to bottom. The display will organize days into 7 columns, corresponding to days of the week, with Sunday on the far left. The week display will show days from left to right, starting with Sunday on the left and Saturday on the far right. In the week display, tasks will be organized column wise chronologically by due date. The day display will display the current day, with tasks organized column wise chronologically by due date. Finally, there will be provided a button for "agenda" which will display all upcoming tasks column wise, organized from top to bottom chronologically by their due dates.

#### **3.100.3 SOURCE**

Customer Meeting

#### **3.100.4 APPLICABLE STANDARDS**

CSS3 font and color standards, WCAG guidelines for accessibility

#### **3.100.5 CONSTRAINTS**

The font sizes and color schemes must comply with accessibility standards to ensure readability. Certain views may not be compatible with all screen sizes.

### **3.101 CALENDAR NAVIGATION**

#### **3.101.1 PRIORITY**

Moderate

### **3.101.2 DESCRIPTION**

Buttons will be provided to select previous or next, which will shift the dates displayed by one increment. In the case of the month display, this will display the previous and next month respectively. In the case of the week display, this will display the previous and next week chronologically respectively. For the day display, this will display the previous and next day chronologically respectively.

### **3.101.3 SOURCE**

Customer Meeting

### **3.101.4 APPLICABLE STANDARDS**

React.js best coding practices

### **3.101.5 CONSTRAINTS**

Only one view may be selected at one time.

## **3.102 ADD TASK ON CLICK OF DAY**

### **3.102.1 PRIORITY**

Moderate

### **3.102.2 DESCRIPTION**

When the user left clicks on a day in any display mode, a subwindow shall be generated. This sub window will display fields for users to input information about a new task. At the bottom of the window, there shall be two buttons, labeled "add task" and "cancel". On the selection of add task, the window will close and the task will be added to the users tasks in the database. On the selection of cancel, the window will close, with no new task generated.

### **3.102.3 SOURCE**

Customer Meeting

### **3.102.4 APPLICABLE STANDARDS**

React.js best coding practices

### **3.102.5 CONSTRAINTS**

New task prompt must have appropriately sized buttons for mobile users

## **3.103 ADD TASK FROM CALENDAR**

### **3.103.1 PRIORITY**

Moderate

### **3.103.2 DESCRIPTION**

The user shall be able to input all relevant information to generate a new task into the sub window. This information shall include; subject (drop down list), task name(string input), priority(drop down list), status(drop down list), start date (calendar), due date (calendar), time due and project (drop down list). The Due Date field will be prefilled with the date that was clicked on.

### **3.103.3 SOURCE**

Customer Meeting

#### **3.103.4 APPLICABLE STANDARDS**

UI/UX confirmation standards

#### **3.103.5 CONSTRAINTS**

The font sizes and color schemes must comply with accessibility standards to ensure readability.

### **3.104 CREATE TASK FROM INPUT**

#### **3.104.1 PRIORITY**

Moderate

#### **3.104.2 DESCRIPTION**

On the click of "add task" the user input to the subwindow shall be added as the fields to a new task and added to the users database profile.

#### **3.104.3 SOURCE**

Customer Meeting

#### **3.104.4 APPLICABLE STANDARDS**

UI/UX confirmation standards. Firestore best coding practices

#### **3.104.5 CONSTRAINTS**

Requires internet access

### **3.105 CALENDAR INITIALIZATION**

#### **3.105.1 PRIORITY**

Low

#### **3.105.2 DESCRIPTION**

On first entry to the calendar page, the display shall be in month mode, and will have the current month displayed.

#### **3.105.3 SOURCE**

Customer Meeting

#### **3.105.4 APPLICABLE STANDARDS**

UI/UX confirmation standards,

#### **3.105.5 CONSTRAINTS**

Month must be applicable to the screen size of the device.

### **3.106 ADD SUBJECT FROM ADD TASK FORM**

#### **3.106.1 PRIORITY**

Moderate

#### **3.106.2 DESCRIPTION**

If "new subject" is selected from the subject drop down in the create task subwindow. A new field will be added to the window. This field will allow users to input the name of a subject, and when "add task" is clicked, a new subject will be created with that name.

### **3.106.3 SOURCE**

Customer Meeting

### **3.106.4 APPLICABLE STANDARDS**

Firestore best coding practices

### **3.106.5 CONSTRAINTS**

Requires internet access

## **3.107 ADD SUBJECT FROM ADD TASK FORM**

### **3.107.1 PRIORITY**

Moderate

### **3.107.2 DESCRIPTION**

If a new project is selected from the subject drop down in the create task subwindow. A new field will be added to the window. This field will allow users to input the name of a project, and when the add task button is clicked, a new project will be created with that name.

### **3.107.3 SOURCE**

Customer Meeting

### **3.107.4 APPLICABLE STANDARDS**

Firestore best coding practices

### **3.107.5 CONSTRAINTS**

Confirm that the project name doesn't already exist. No duplicate project names will be allowed, requires internet access

## **3.108 EDIT AND DELETE TASKS FROM CALENDAR VIEW**

### **3.108.1 PRIORITY**

Moderate

### **3.108.2 DESCRIPTION**

if a user left clicks on a task, a subwindow will be generated that shall display all task information and have button options to edit, delete or close. If edit is clicked, a new subwindow will be generated, that displays the task information and allows the user to edit each field. If delete is clicked, the task will be deleted from the database. If close is clicked, the subwindow will close.

### **3.108.3 SOURCE**

Customer Meeting

### **3.108.4 APPLICABLE STANDARDS**

UI/UX confirmation standards, firebase data persistence standards.

### **3.108.5 CONSTRAINTS**

Requires internet access

### **3.109 CALENDAR FORMAT OF CURRENT DAY (MOBILE)**

#### **3.109.1 PRIORITY**

Moderate

#### **3.109.2 DESCRIPTION**

The system shall display the current date with a light blue background()

#### **3.109.3 SOURCE**

Customer Meeting

#### **3.109.4 APPLICABLE STANDARDS**

UI/UX confirmation standards, WCAG guidelines for accessibility

#### **3.109.5 CONSTRAINTS**

All specified colors must meet WCAG standards for accessibility, particularly in terms of contrast ratios.

### **3.110 CALENDAR FORMAT OF OUT-OF-RANGE DATES (MOBILE)**

#### **3.110.1 PRIORITY**

Moderate

#### **3.110.2 DESCRIPTION**

Dates not in the current month shall be displayed with a dark gray () background if they are part of a week that is currently displayed.

#### **3.110.3 SOURCE**

Customer Meeting

#### **3.110.4 APPLICABLE STANDARDS**

UI/UX confirmation standards

#### **3.110.5 CONSTRAINTS**

All specified colors must meet WCAG standards for accessibility, particularly in terms of contrast ratios.

## 4 PACKAGING REQUIREMENTS

Include a header paragraph here. Packaging requirements are those requirements that identify how the delivered product will be packaged for delivery to the end-user; or how it will "look" when finished and delivered. For example, you might specify that the software required for operation will be pre-loaded on the hard drive, delivered on CD/DVD, or available via download. Software might be customer installable, or not, etc. Hardware components could be all in a single package, provided as a "bag of parts" to be assembled/installed by the user, painted a certain color, logos affixed, etc. Care should be taken not to duplicate requirements found in other sections of this document.

### 4.1 PROGRESSIVE WEB APPLICATION (PWA) DELIVERY

#### 4.1.1 DESCRIPTION

The platform shall be accessible via a web browser with an option to install as a PWA, allowing users to use it as a mobile or desktop app without needing to download it from an app store.

#### 4.1.2 SOURCE

LearnLeaf Organizer Website

#### 4.1.3 CONSTRAINTS

The PWA must be installable across different platforms (Windows, iOS, Android) and follow the Web App Manifest standards to allow easy installation.

#### 4.1.4 STANDARDS

W3C Web App Manifest, Service Worker API

#### 4.1.5 PRIORITY

Critical

### 4.2 DIRECT DOWNLOAD AND ACCESS

#### 4.2.1 DESCRIPTION

Users shall access the LearnLeaf Organizer through a secure HTTPS URL. There shall be no need for users to download traditional software components, as the PWA will be accessible entirely via the web.

#### 4.2.2 SOURCE

LearnLeaf Organizer Website

#### 4.2.3 CONSTRAINTS

The URL must support SSL certificates (HTTPS), ensuring secure communication and user authentication.

#### 4.2.4 STANDARDS

SSL/TLS Certificates, HTTPS Protocol Standards (RFC 2818)

#### 4.2.5 PRIORITY

High

### 4.3 FIREBASE PRE-CONFIGURED

#### **4.3.1 DESCRIPTION**

The platform shall be configured to use Firebase Firestore for database management and Firebase Authentication for user management. These services shall be fully integrated to allow seamless real-time data storage and user authentication.

#### **4.3.2 SOURCE**

LearnLeaf Organizer Website

#### **4.3.3 CONSTRAINTS**

The back-end services (Firebase Firestore, Firebase Authentication) must be fully integrated, ensuring smooth operation and minimal configuration for the user or administrator. The platform must not rely on Firebase Hosting for deployment.

#### **4.3.4 STANDARDS**

Firebase Firestore API, Firebase Authentication Standards

#### **4.3.5 PRIORITY**

High

### **4.4 NO THIRD-PARTY SOFTWARE FOR INSTALLATION OR USE**

#### **4.4.1 DESCRIPTION**

No additional software (such as plug-ins or client-side libraries) shall be required for users to access and use the LearnLeaf Organizer platform.

#### **4.4.2 SOURCE**

LearnLeaf Organizer Website

#### **4.4.3 CONSTRAINTS**

The platform must operate entirely within the web browser and rely on modern web standards (such as HTML5, CSS3, and JavaScript).

#### **4.4.4 STANDARDS**

HTML5, CSS3, JavaScript (ES6+)

#### **4.4.5 PRIORITY**

High

### **4.5 APK PACKAGE (ANDROID) DELIVERY**

#### **4.5.1 DESCRIPTION**

The mobile version of LearnLeaf Organizer shall be packaged as an APK file for installation on Android devices.

#### **4.5.2 SOURCE**

LearnLeaf Organizer Website

#### **4.5.3 CONSTRAINTS**

The APK must be signed using a valid developer key to ensure integrity and security.

#### **4.5.4 STANDARDS**

APK Signing Guidelines (Google Play), Android APK Packaging Standards

#### **4.5.5 PRIORITY**

High

### **4.6 MOBILE VERSION TO BE DISTRIBUTED VIA GOOGLE PLAY STORE (ANDROID)**

#### **4.6.1 DESCRIPTION**

The mobile app shall be submitted to and distributed via the Google Play Store to ensure ease of access for Android users.

#### **4.6.2 SOURCE**

LearnLeaf Organizer Website

#### **4.6.3 CONSTRAINTS**

The mobile app must comply with Google Play Store guidelines, including policies on app permissions, security, and updates.

#### **4.6.4 STANDARDS**

Google Play Store Guidelines, Android App Security Standards

#### **4.6.5 PRIORITY**

Critical

## 5 PERFORMANCE REQUIREMENTS

Include a header paragraph specific to your product here. Performance requirements address items such as: how fast specific critical operations must complete; how long it takes to start/stop activities; how long the battery must last; maximum time it must take to set up; etc.

### 5.1 OPTIMIZED LOADING TIME

#### 5.1.1 DESCRIPTION

The system shall implement an App Shell architecture that quickly loads the core interface for a smooth user experience.

#### 5.1.2 SOURCE

Customer Meeting

#### 5.1.3 CONSTRAINTS

The app shell must be lightweight and load instantly, fetching dynamic content later as the network allows.

#### 5.1.4 STANDARDS

PWA App Shell design standards

#### 5.1.5 PRIORITY

Low

### 5.2 FAST LOADING ON SLOW NETWORKS

#### 5.2.1 DESCRIPTION

The system shall ensure the initial load time is under 3 seconds on 3G networks, optimizing asset delivery.

#### 5.2.2 SOURCE

Customer Meeting

#### 5.2.3 CONSTRAINTS

Assets should be minified and compressed; lazy loading should be implemented for non-critical content.

#### 5.2.4 STANDARDS

Google Lighthouse PWA performance guidelines, Web Performance Optimization

#### 5.2.5 PRIORITY

Low

### 5.3 EFFICIENT MEMORY USAGE

#### 5.3.1 DESCRIPTION

When creating notification documents, Firestore is queried for tasks based on user preferences, filtering out completed tasks. Queries must be efficient for large datasets.

#### 5.3.2 SOURCE

Customer Meeting

### **5.3.3 CONSTRAINTS**

Queries must minimize latency and optimize performance for users with a large number of tasks.

### **5.3.4 STANDARDS**

Firebase Firestore best practices, Query optimization standards

### **5.3.5 PRIORITY**

High

## **5.4 LOW LATENCY FOR REAL-TIME UPDATES**

### **5.4.1 DESCRIPTION**

The dashboard performance shall be optimized, especially with large datasets, using techniques like lazy loading.

### **5.4.2 SOURCE**

Customer Meeting

### **5.4.3 CONSTRAINTS**

Must manage large datasets efficiently.

### **5.4.4 STANDARDS**

React.js, lazy loading

### **5.4.5 PRIORITY**

High

## **5.5 BACKGROUND DATA SYNC**

### **5.5.1 DESCRIPTION**

The system shall use background sync to update user data when the app is opened after being offline.

### **5.5.2 SOURCE**

Customer Meeting

### **5.5.3 CONSTRAINTS**

Syncing must occur efficiently without draining device resources or degrading app performance.

### **5.5.4 STANDARDS**

Background Sync API

### **5.5.5 PRIORITY**

Future

## **5.6 OFFLINE MODE**

### **5.6.1 SOURCE**

Benjamin Niccum

### **5.6.2 DESCRIPTION**

The app shall provide offline functionality, allowing users to view and edit tasks without an internet connection.

### **5.6.3 REQUIREMENTS AREA**

Performance

### **5.6.4 IMPORTANCE**

High

### **5.6.5 APPLICABLE STANDARD**

IndexedDB, SQLite

### **5.6.6 CONSTRAINTS**

Data must be synced when the connection is restored.

## **5.7 BACKGROUND SYNC**

### **5.7.1 SOURCE**

Benjamin Niccum

### **5.7.2 DESCRIPTION**

The app shall perform background syncing of tasks and user data to ensure up-to-date information.

### **5.7.3 REQUIREMENTS AREA**

Performance

### **5.7.4 IMPORTANCE**

Critical

### **5.7.5 APPLICABLE STANDARD**

Android WorkManager, iOS Background Fetch

### **5.7.6 CONSTRAINTS**

Optimize for battery life and respect system limitations.

## **5.8 FETCH ARCHIVED DATA**

### **5.8.1 SOURCE**

Database Administrator

### **5.8.2 DESCRIPTION**

The system shall fetch archived tasks, subjects, and projects associated with the logged-in user.

### **5.8.3 REQUIREMENTS AREA**

Performance

### **5.8.4 IMPORTANCE**

High

### **5.8.5 APPLICABLE STANDARD**

SQL, NoSQL database standards

### **5.8.6 CONSTRAINTS**

Optimize database queries for efficient data retrieval.

## **5.9 TASK CREATION IN FIREBASE**

### **5.9.1 SOURCE**

Learnleaf Organizer Website

### **5.9.2 DESCRIPTION**

The system shall pass the new task details to Firebase to create a document in the Tasks collection in Firestore Database. Upon task creation, the system shall add the task details to Firebase's Firestore Database as a new document in the Tasks collection.

### **5.9.3 REQUIREMENTS AREA**

Performance

### **5.9.4 IMPORTANCE**

High

### **5.9.5 APPLICABLE STANDARD**

Firebase Firestore data handling standards

### **5.9.6 CONSTRAINTS**

Task data types must align with Firebase's data requirements for seamless storage.

## **5.10 SUBJECT CREATION IN FIREBASE**

### **5.10.1 SOURCE**

Learnleaf Organizer Website

### **5.10.2 DESCRIPTION**

The system shall pass the new subject details to Firebase to create a document in the Subjects collection in Firestore Database. Upon subject creation, the system shall add the subject details to Firebase's Firestore Database as a new document in the Subjects collection.

### **5.10.3 REQUIREMENTS AREA**

Performance

### **5.10.4 IMPORTANCE**

High

### **5.10.5 APPLICABLE STANDARD**

Firebase Firestore data handling standards

### **5.10.6 CONSTRAINTS**

Task data types must align with Firebase's data requirements for seamless storage.

## **5.11 PROJECT CREATION IN FIREBASE**

### **5.11.1 SOURCE**

Learnleaf Organizer Website

#### **5.11.2 DESCRIPTION**

The system shall pass the new project details to Firebase to create a document in the Projects collection in Firestore Database. Upon project creation, the system shall add the project details to Firebase's Firestore Database as a new document in the Projects collection.

#### **5.11.3 REQUIREMENTS AREA**

Performance

#### **5.11.4 IMPORTANCE**

High

#### **5.11.5 APPLICABLE STANDARD**

Firebase Firestore data handling standards

#### **5.11.6 CONSTRAINTS**

Task data types must align with Firebase's data requirements for seamless storage.

## **6 SAFETY REQUIREMENTS**

LearnLeaf Organizer is a web-based platform that does not involve the use of any physical hardware or hazardous materials. The platform is purely software-based, designed to be accessed via a web browser or mobile app. Therefore, no specific safety requirements related to user health or physical harm are necessary. The platform does not expose users to any potential dangers such as electrical hazards, toxic materials, or physical risks commonly associated with hardware products.

## 7 SECURITY REQUIREMENTS

Include a header paragraph specific to your product here. In this section specify any requirements related to information security or privacy. This may include items such as encryption standards, data storage procedures, authentication, password strength, etc.

### 7.1 HTTPS ENFORCEMENT

#### 7.1.1 DESCRIPTION

The system shall enforce HTTPS for all communication, securing data transmission between users and the server.

#### 7.1.2 SOURCE

Customer Meeting

#### 7.1.3 CONSTRAINTS

All communication must be over HTTPS, with regular certificate maintenance and updates.

#### 7.1.4 STANDARDS

SSL/TLS encryption, HTTPS requirement for PWAs

#### 7.1.5 PRIORITY

Critical

### 7.2 EMAIL NOTIFICATION SECURITY

#### 7.2.1 DESCRIPTION

Emails shall be sent from the address 'learnleaforganizer@gmail.com', which must be verified with SendGrid to avoid spam issues.

#### 7.2.2 SOURCE

Customer Meeting

#### 7.2.3 CONSTRAINTS

Ensure proper authentication using SPF and DKIM to avoid being flagged as spam.

#### 7.2.4 STANDARDS

SendGrid verification requirements, SPF, DKIM, CAN-SPAM Act

#### 7.2.5 PRIORITY

Critical

### 7.3 COMPLIANCE WITH BULK EMAIL STANDARDS

#### 7.3.1 DESCRIPTION

Bulk email sending must comply with Gmail and SendGrid guidelines, including verifying sender identity, email quotas, and email rate limits. Emails must use proper authentication, such as SPF and DKIM, to avoid being marked as spam. Unsubscribe links must be included in compliance with email laws like the CAN-SPAM Act.

#### 7.3.2 SOURCE

Customer Meeting

### **7.3.3 CONSTRAINTS**

Bulk sending must comply with rate limits (e.g., 2,000 emails/day for Gmail). Emails must include a clear "Unsubscribe" option to comply with CAN-SPAM regulations.

### **7.3.4 STANDARDS**

Gmail bulk email standards, SendGrid bulk sending guidelines, CAN-SPAM Act

### **7.3.5 PRIORITY**

Critical

## **7.4 SECURE LOGIN FORM**

### **7.4.1 SOURCE**

Benjamin Niccum

### **7.4.2 DESCRIPTION**

The system shall provide a secure login form with fields for username/email and password.

### **7.4.3 REQUIREMENTS AREA**

Security

### **7.4.4 IMPORTANCE**

Critical

### **7.4.5 APPLICABLE STANDARD**

OWASP Authentication Guidelines

### **7.4.6 CONSTRAINTS**

Form must use HTTPS and implement CSRF protection.

## **7.5 PASSWORD MASKING**

### **7.5.1 SOURCE**

Benjamin Niccum

### **7.5.2 DESCRIPTION**

The system shall mask password input, with an option to toggle visibility.

### **7.5.3 REQUIREMENTS AREA**

Security

### **7.5.4 IMPORTANCE**

High

### **7.5.5 APPLICABLE STANDARD**

Web Content Accessibility Guidelines (WCAG) 2.1

### **7.5.6 CONSTRAINTS**

Ensure toggle is accessible via keyboard and screen readers.

## **7.6 PASSWORD RESET OPTION**

### **7.6.1 SOURCE**

Customer Meeting

### **7.6.2 DESCRIPTION**

The system shall provide a "Forgot Password" link to initiate password reset.

### **7.6.3 REQUIREMENTS AREA**

Security

### **7.6.4 IMPORTANCE**

High

### **7.6.5 APPLICABLE STANDARD**

NIST Digital Identity Guidelines

### **7.6.6 CONSTRAINTS**

Implement rate limiting and secure reset token generation.

## **7.7 BRUTE FORCE PROTECTION**

### **7.7.1 SOURCE**

Customer Meeting

### **7.7.2 DESCRIPTION**

The system shall implement measures to prevent brute force attacks.

### **7.7.3 REQUIREMENTS AREA**

Security

### **7.7.4 IMPORTANCE**

Critical

### **7.7.5 APPLICABLE STANDARD**

OWASP Brute Force Prevention Cheat Sheet

### **7.7.6 CONSTRAINTS**

Implement account lockouts and CAPTCHAs after multiple failed attempts.

## **7.8 LOGIN ACTIVITY LOGGING**

### **7.8.1 SOURCE**

Customer Meeting

### **7.8.2 DESCRIPTION**

The system shall log all login attempts for security auditing.

### **7.8.3 REQUIREMENTS AREA**

Security

### **7.8.4 IMPORTANCE**

High

#### **7.8.5 APPLICABLE STANDARD**

GDPR, CCPA compliance

#### **7.8.6 CONSTRAINTS**

Ensure logs are securely stored and do not contain sensitive information.

### **7.9 PASSWORD STRENGTH INDICATOR**

#### **7.9.1 SOURCE**

Customer Meeting

#### **7.9.2 DESCRIPTION**

The system shall display a password strength indicator during registration.

#### **7.9.3 REQUIREMENTS AREA**

Security

#### **7.9.4 IMPORTANCE**

Moderate

#### **7.9.5 APPLICABLE STANDARD**

NIST Password Guidelines

#### **7.9.6 CONSTRAINTS**

Implement real-time password strength checking without sending password to server.

### **7.10 EMAIL VERIFICATION**

#### **7.10.1 SOURCE**

Benjamin Niccum

#### **7.10.2 DESCRIPTION**

The system shall require email verification to complete registration.

#### **7.10.3 REQUIREMENTS AREA**

Security

#### **7.10.4 IMPORTANCE**

Critical

#### **7.10.5 APPLICABLE STANDARD**

Anti-Spam best practices

#### **7.10.6 CONSTRAINTS**

Implement secure token generation and expiration for verification links.

### **7.11 CAPTCHA IMPLEMENTATION**

#### **7.11.1 SOURCE**

Customer Meeting

#### **7.11.2 DESCRIPTION**

The system shall implement CAPTCHA to prevent automated registrations.

### **7.11.3 REQUIREMENTS AREA**

Security

### **7.11.4 IMPORTANCE**

Moderate

### **7.11.5 APPLICABLE STANDARD**

reCAPTCHA v3 or hCaptcha

### **7.11.6 CONSTRAINTS**

Balance security with user experience, consider accessibility implications.

## **7.12 USER AUTHENTICATION AND LOGOUT**

### **7.12.1 SOURCE**

System Administrator

### **7.12.2 DESCRIPTION**

The system shall authenticate the user and allow them to log out.

### **7.12.3 REQUIREMENTS AREA**

Security

### **7.12.4 IMPORTANCE**

Critical

### **7.12.5 APPLICABLE STANDARD**

OAuth 2.0, OpenID Connect

### **7.12.6 CONSTRAINTS**

Implement secure authentication protocols and ensure user data privacy.

## 8 MAINTENANCE & SUPPORT REQUIREMENTS

Include a header paragraph specific to your product here. Maintenance and support requirements address items specific to the ongoing maintenance and support of your product after delivery. Think of these requirements as if you were the ones who would be responsible for caring for customers/end user after the product is delivered in its final form and in use "in the field". What would you require to do this job? Specify items such as: where, how and who must be able to maintain the product to correct errors, hardware failures, etc.; required support/troubleshooting manuals/guides; availability/documentation of source code; related technical documentation that must be available for maintainers; specific/unique tools required for maintenance; specific software/environment required for maintenance; etc.

### 8.1 PERFORMANCE AUDITING TOOLS

#### 8.1.1 DESCRIPTION

The system shall be audited for performance regularly using tools like Google Lighthouse.

#### 8.1.2 SOURCE

Customer Meeting

#### 8.1.3 CONSTRAINTS

The app must progressively enhance features based on the user's device and network capabilities.

#### 8.1.4 STANDARDS

Google Lighthouse performance auditing tools

#### 8.1.5 PRIORITY

Moderate

### 8.2 VERSION CONTROL SUPPORT

#### 8.2.1 DESCRIPTION

The system shall automatically update the PWA when new versions are deployed, ensuring that users always have the latest features and fixes without manual intervention. This should be handled via the service worker, which checks for updates in the background and applies them when the app is next opened.

#### 8.2.2 SOURCE

Customer Meeting

#### 8.2.3 CONSTRAINTS

Updates must be applied seamlessly without disrupting the user's workflow. If the PWA is open during the update, users should be prompted to refresh to apply the new version. The update process must ensure data integrity and avoid conflicts with cached data.

#### 8.2.4 STANDARDS

W3C Service Worker API, PWA Update Guidelines

#### 8.2.5 PRIORITY

High

## **8.3 NOTIFICATION STORAGE**

### **8.3.1 DESCRIPTION**

Notification documents shall be created and stored in Firestore, containing user ID, email, task details, notification type, and timestamp.

### **8.3.2 SOURCE**

Customer Meeting

### **8.3.3 CONSTRAINTS**

Documents must be structured efficiently to allow quick retrieval and email generation.

### **8.3.4 STANDARDS**

Firestore database structure guidelines, Firebase Admin SDK

### **8.3.5 PRIORITY**

High

## 9 OTHER REQUIREMENTS

Include a header paragraph specific to your product here. In this section specify anything else that is required for the product to be deemed complete. Include requirements related to customer setup and configuration if not specified in a previous requirement. Add any known requirements related to product architecture/design, such as modularity, extensibility (for future enhancements), or adaptation for a specific programming language. Consider requirements such as portability of your source code to various platforms (Windows, Linux, Unix Mac OS, etc.).

### 9.1 PWA MANIFEST FILE

#### 9.1.1 DESCRIPTION

The system shall include a manifest file with icons, app name, theme color, and display mode for installation.

#### 9.1.2 SOURCE

Customer Meeting

#### 9.1.3 CONSTRAINTS

The manifest file must include proper icons, names, and colors in compliance with branding guidelines.

#### 9.1.4 STANDARDS

W3C Web App Manifest Specification

#### 9.1.5 PRIORITY

High

## **10 FUTURE ITEMS**

In this last section, you will reiterate all requirements that are listed as priority 5. This is repetitive, but necessary as a concise statement of features/functions that were considered/discussed and documented herein, but will NOT be addressed in the prototype version of the product due to constraints of budget, time, skills, technology, feasibility analysis, etc. Use the following format for this section.

### **10.1 BACKGROUND DATA SYNC**

#### **10.1.1 DESCRIPTION**

The system shall use background sync to update user data when the app is opened after being offline.

#### **10.1.2 SOURCE**

Customer Meeting

#### **10.1.3 CONSTRAINTS**

Syncing must occur efficiently without draining device resources or degrading app performance.

#### **10.1.4 STANDARDS**

Background Sync API

#### **10.1.5 PRIORITY**

Future

### **10.2 SOCIAL MEDIA LOGIN INTEGRATION**

#### **10.2.1 SOURCE**

Customer Meeting

#### **10.2.2 DESCRIPTION**

The system shall allow login via popular social media platforms.

#### **10.2.3 REQUIREMENTS AREA**

Future

#### **10.2.4 IMPORTANCE**

Moderate

#### **10.2.5 APPLICABLE STANDARD**

OAuth 2.0, OpenID Connect

#### **10.2.6 CONSTRAINTS**

Comply with each platform's authentication policies and data usage guidelines.

### **10.3 WIDGET SUPPORT**

#### **10.3.1 SOURCE**

Customer Meeting

#### **10.3.2 DESCRIPTION**

The app shall provide home screen widgets for quick task viewing and management.

### **10.3.3 REQUIREMENTS AREA**

Future

### **10.3.4 IMPORTANCE**

High

### **10.3.5 APPLICABLE STANDARD**

Android App Widgets, iOS

### **10.3.6 CONSTRAINTS**

Widgets must be responsive and update in real-time.

## **10.4 VOICE INPUT FOR TASKS**

### **10.4.1 SOURCE**

Customer Meeting

### **10.4.2 DESCRIPTION**

The app shall allow users to create tasks and set reminders using voice input.

### **10.4.3 REQUIREMENTS AREA**

Future

### **10.4.4 IMPORTANCE**

High

### **10.4.5 APPLICABLE STANDARD**

Android Speech Recognizer, iOS Speech framework

### **10.4.6 CONSTRAINTS**

Handle multiple languages and accents effectively.

## REFERENCES

- [1] OpenAI. Chatgpt, (2024).